A PROJECT
REPORT ON

Dance and Music Website

Aadhiya Shakti Academy

Dance • Singing • Music • Choreography

SUBMITTED BY

Ms. Tamanna N. Dama (24CS002UG04215)
Mr. Harshal C. Mehta (24CS002UG04293)

ACADEMIC YEAR 2026-27 T.Y. B.C.A. SEM-5

UNDER THE GUIDANCE OF

Prof. Khushal Rajani

SMT. C.Z.M. GOSRANI BCA & M.SC. (IT & CA) COLLEGE – JAMNAGAR

SUBMITTED TO

SAURASHTRA UNIVERSITY – RAJKOT

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CZMG BCA-135/2026-27/152 Date: 01/10/2026

CERTIFICATE OF PROJECT COMPLETION

This is to certify that the project entitled
"Dance and Music Website"
is a bonafide work carried out by
Tamanna Dama N. (24CS002UG04215)
and **Harshal Mehta C. (24CS002UG04293)**
as a partial fulfillment of the requirements
for the B.C.A. Semester 5 during the
Academic Year 2026-2027.

The project report meets the academic requirements for the project work prescribed by Saurashtra University, Rajkot.

EXAMINER NAME & SIGN	**PRINCIPAL SIGN**
Mr. Khushal Rajani	/ /2026
PROJECT GUIDE	**EXAM DATE**



ABSTRACT

The **Dance and Music Website** – **Aadhiya Shakti Academy** – is a dynamic, database-driven website developed for a dance, singing, music and choreography academy in Jamnagar, Gujarat. The project was assigned by Smt. C.Z.M. Gosrani B.C.A. College as part of the T.Y. B.C.A. Semester 5 curriculum. Its aim is to give the academy a professional online presence and, at the same time, a secure admin portal through which the academy can manage its own content without writing code.

The public website has eleven pages – Home, About, Courses, Course Detail, Instructors, Instructor Detail, Gallery, Events, Services, Testimonials and Contact. Visitors can browse 32 courses in three categories (Dance, Singing and Choreography), read tutor profiles, view photographs and events, and send an enquiry. Every enquiry is validated on the server and stored in a MySQL database. The design uses the academy's plum, rose, pink and cream palette with smooth scroll, parallax and cursor-glow animations.

The Admin Portal is protected by login and a role-based permission system. A Super Admin can create other admins (Admin, Content Manager, Course Manager, Tutor Manager), customise their permissions and activate or deactivate them. Admins can manage courses, categories, subcategories, tutors, photos and enquiries, upload images with live preview and review an activity log. Logout ends the session and returns the user directly to the public home page.

Key features of the system:

- Eleven-page responsive public website with a consistent visual identity
- Contact enquiry form with server-side validation, stored in MySQL
- Secure login – password hashing, CSRF tokens and session regeneration
- Role-based access control enforced on the server for every action
- Full CRUD for courses, categories, subcategories and tutors
- Secure image upload with real-content validation and live preview
- Dashboard, activity log, search, filters, confirmation dialogs and notifications
- No online payment – prices are shown for information only

The system is built with PHP 8, MySQL (PDO with prepared statements), HTML5, CSS3 and JavaScript, and runs on Apache and MySQL through MAMP. Its modular structure makes future additions, such as a gallery manager and e-mail confirmation, simple to integrate.



ACKNOWLEDGEMENT

We take immense pleasure in submitting this project report titled **"Dance and Music Website"** (Aadhiya Shakti Academy) as a part of the T.Y. B.C.A. Semester 5 curriculum of Saurashtra University. This project gave us valuable practical experience in planning, designing, developing, testing and documenting a complete database-driven web application.

We are deeply grateful to our Project Guide, **Prof. Khushal Rajani Sir**, for his continuous guidance, valuable suggestions and constructive feedback at every stage of the project. His encouragement helped us understand the subject clearly and complete the work with confidence.

We express our sincere gratitude to our Principal, **Dr. (Mrs.) Hetal Savla Ma'am**, and to all the faculty members of **Smt. C.Z.M. Gosrani B.C.A. & M.Sc. (IT & CA) College, Jamnagar**, for giving us this opportunity and for providing the academic support, laboratory facilities and resources required for the project.

We are also thankful to our parents and friends for their constant support, patience and motivation throughout the development of this project. Finally, we thank everyone who directly or indirectly helped us in completing this project successfully.

Ms. Tamanna N. Dama
Mr. Harshal C. Mehta



PROJECT PROFILE

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Enrollment Number	24CS002UG04215
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PROJECT DETAILS	
Project Title	Dance and Music Website
Website Name	Aadhiya Shakti Academy
Project Period	August – October 2026
Type of Project	Dynamic Web Application (Online)
Platform	PHP 8, MySQL, HTML5, CSS3, JavaScript, Apache (MAMP)
Operating System	macOS (development) – Windows / Linux compatible
Team Size	2 Members
GUIDE INFORMATION	
Name of Guide	Prof. Khushal Rajani
College	Smt. C.Z.M. Gosrani BCA & M.Sc. (IT & CA) College, Jamnagar
University	Saurashtra University, Rajkot

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CHAPTER 1

SDLC OVERVIEW



Figure 1: Software Development Life Cycle followed for Aadhiya Shakti Academy

The project followed the classical Software Development Life Cycle (SDLC) in six steps. Each step produced a concrete output that was reviewed before moving to the next one.



Step 1 – Planning

The academy's goals were written down: present the dance, singing, music and choreography services; collect enquiries; let the owner manage content without programming. The technology (PHP, MySQL, Apache through MAMP) and the work phases were selected.

Step 2 – Designing

The visual identity (plum #601D49, rose #BD5579, pink #EA9D9D, cream #FFEBB8), page layouts, navigation, the database structure and the admin sidebar were designed before coding started.

Step 3 – Development

Eleven public pages, the contact form, the admin login, course / category / tutor management, member roles, image upload, activity log and the responsive admin dashboard were coded in small phases.

Step 4 – Testing

Login and logout, role permissions, direct-request attacks, upload validation, duplicate data, search filters and every public page were tested (see Chapter 5).

Step 5 – Deployment

The project is deployed by importing one SQL file (full-setup.sql) in phpMyAdmin and copying the folder into the MAMP htdocs directory (see Appendix A).

Step 6 – Maintenance

Courses, tutors and photos are updated by the admin through the portal. Future work: gallery management, e-mail confirmation and more page animations.

CHAPTER 2

REQUIREMENT GATHERING AND ANALYSIS

Requirement gathering is the first and most important step of the project. In this phase the needs of the project owner and of the end users were collected, discussed and analysed so that the final system would solve the right problems. The details are given below.

2.1 ORGANIZATION DETAILS

1. Name of the Organization:

Smt. C.Z.M. Gosrani B.C.A. & M.Sc. (IT & CA) College (CZMG College), Jamnagar – affiliated to Saurashtra University, Rajkot

2. Brief detail of the organization:

Smt. C.Z.M. Gosrani B.C.A. & M.Sc. (IT & CA) College, Jamnagar, provides education and practical training to students of computer applications. As part of the T.Y. B.C.A. Semester 5 curriculum, the college assigned this project to our team under the guidance of Prof. Khushal Rajani. **The college is therefore the owner of this project.**

The project is a complete website for **Aadhiya Shakti Academy**, a dance, singing, music and choreography academy in Jamnagar, Gujarat (established 2017). The academy offers training in classical, folk, Garba, Bollywood and contemporary dance, in vocal styles, and event choreography for weddings, schools, colleges, offices and cultural functions. The website must present these services, collect enquiries and let the administrator manage all content easily.

2.2 MEETINGS / REQUIREMENT DISCUSSIONS

Two types of discussions were held: one with the project owner (CZMG College) to understand the scope and expected features, and one with the intended users of the website to understand what they expect when they visit an academy website.

I. Meeting with the Project Owner

1. Name of Owner:

CZMG College – Smt. C.Z.M. Gosrani B.C.A. College, Jamnagar (represented by Prof. Khushal Rajani, Project Guide)

2. Owner's requirements:



The college explained that the project should be a real-world, database-driven website that demonstrates front-end design, back-end programming, database design and security. The main requirements given were:

- The website must present the academy's dance, singing, music and choreography services in the academy colours – plum, rose and cream.
- Every Contact form submission must be stored in a database and shown to the administrator on a separate admin page.
- The administrator must be able to change course prices, upload photos from a computer, and add, edit or remove courses and tutors.
- The main administrator must be able to create other admin accounts, each with only the rights it needs.
- On logout the administrator must return to the public home page.
- No online payment system is to be included.

II. Meeting with the User (prospective student / parent)

1. Name of the user:

Prospective students and parents (website visitors)

2. User requirements:

Users want to find information quickly and contact the academy without difficulty. Their main expectations were:

- Find a course easily and see its level, duration, mode and price.
- Know which tutor will teach the course and read the tutor's profile.
- See photographs of performances, events and classes.
- Send an enquiry easily and be sure it has been received.
- Use the website comfortably on a mobile phone as well as a laptop.

2.3 DATA WHICH WILL BE THE INPUT INTO THE SYSTEM

The system accepts the following data from visitors and administrators. All input is validated on the server before it is saved.

- **Text** – contact enquiries, course details, tutor profiles and admin account details
- **Images** – course images, tutor photos and site photos (JPG, PNG, WEBP or GIF, maximum 5 MB)



- **Selections** – category, subcategory, level, mode, role, permissions and status
- **Credentials** – admin username and password

2.4 DATA WHICH WILL BE THE OUTPUT FROM THE SYSTEM

The processed data is shown to visitors on the public pages and to administrators in the Admin Portal:

- Public pages – courses, tutors, gallery, events, services and testimonials
- Enquiry list with status (new / read / archived)
- Dashboard statistics and courses-by-category chart
- Activity log of all important admin actions
- Success notifications and validation error messages

2.5 TYPE OF PROJECT

This is an **online, dynamic web application** based on the client–server model. It follows a three-tier architecture: the browser (presentation layer), PHP on Apache (application layer) and MySQL (data layer).

2.6 METHOD FOR COLLECTING DATA

The following methods were used to collect and confirm the requirements:

- **Interview / discussion** with the project guide and the written project brief given by the college
- **Observation** of existing dance and music academy websites
- **Content study** of the academy's courses, categories and services
- **Iterative feedback** – each phase was shown, tested and corrected before the next one

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATIONS

3.1 Introduction

3.1.1 Purpose

The purpose of this System Requirement Specification is to describe clearly what the Aadhiya Shakti Academy website must do and the conditions under which it must work. The system presents the academy online, collects enquiries from visitors and provides a secure Admin Portal for managing courses, categories, tutors, photos, messages and administrator accounts.

3.1.2 Intended Audience and Reading Suggestions

This document is intended for the project guide, the examiners, the academy administrator and any future developer. It is recommended to read the chapters in order: Chapter 3 explains what the system does, Chapter 4 explains how the data is organised, Chapter 5 shows the testing and Chapter 6 shows the actual screens.

3.1.3 Product Scope

The system covers three main areas: the public information website, the contact enquiry system and the Admin Portal with role-based access. Online payment, live class streaming and student login are outside the scope of this version and are listed as future enhancements.

3.2 Overall Description

3.2.1 Product Function

The system is divided into two modules. The public module is used by visitors, and the admin module is used by authorised administrators.

(i) Public Website Module

- Home, About, Courses (with search and category filter) and Course Detail
- Instructors and Instructor Detail
- Gallery, Events, Services and Testimonials
- Contact form with service selection and Service Areas panel

(ii) Admin Module



- Login, Logout and My Password
- Dashboard with statistics, category chart, latest enquiries and recent activity
- Courses, Categories and Subcategories management
- Tutors management with image upload and safe delete
- Messages (view, mark read, archive, delete) and Site Photos
- Admin Members with roles and permissions, and the Activity Log

3.2.2 User Classes and Characteristics

The system has one public user class and five administrator roles. Each role sees only the menu items and actions allowed by its permissions.

Table 1: User classes

User class	Characteristics
Visitor / Student	Public user; can view pages and send an enquiry. No login required.
Super Admin	Full control; manages all content and all other admins. The last active Super Admin cannot be removed.
Admin	Manages courses, categories, tutors, photos and messages; cannot manage admin members.
Content Manager	Manages site photos, course and tutor images and messages.
Course Manager	Manages courses, course images and categories.
Tutor Manager	Manages tutors and tutor images.

3.2.3 Operating Environment

The system is developed with PHP 8, MySQL, HTML5, CSS3 and JavaScript. It runs on the MAMP stack (Apache web server and MySQL database) on macOS, and can run equally on Windows or Linux using XAMPP or LAMP. The website is accessed through any modern browser such as Google Chrome, Safari, Microsoft Edge or Mozilla Firefox, on both desktop and mobile devices.

3.2.4 Design and Implementation Constraints

The following constraints were followed during design and development:

- Developed with PHP, MySQL (PDO), HTML, CSS and JavaScript only – no heavy frameworks
- Requires Apache and MySQL (MAMP default: MySQL port 8889, user root, password root)



- Uploaded images are limited to JPG, PNG, WEBP and GIF, up to 5 MB
- Only authorised and active admins can access the Admin Portal
- No online payment is implemented

3.2.5 Assumptions and Dependencies

The correct working of the system depends on the following assumptions:

- MAMP (Apache and MySQL) is running and the database has been imported
- An internet connection is available for Google Fonts and hosted photographs
- Administrators enter correct information and keep their passwords secret
- Placeholder contact details are replaced with the academy's real details before going live

3.3 External Interface Requirements

3.3.1 User Interface

The interface is clean, responsive and consistent, using the academy's plum, rose, pink and cream colours. The public site has a top navigation bar, animated course and tutor cards and a common footer. The Admin Portal has a dark-plum collapsible sidebar with icons, statistic cards, tables with search and filters, forms with image preview, confirmation dialogs before deletion and short toast notifications after every action.

3.3.2 Software Interface

- Operating System: macOS / Windows / Linux
- Web Server: Apache (MAMP / XAMPP)
- Back end: PHP 8 (strict types, PDO)
- Database: MySQL / MariaDB, managed with phpMyAdmin
- Front end: HTML5, CSS3 and JavaScript (no external animation library)
- Browser: Google Chrome (tested), Safari, Edge, Firefox
- Fonts: Fraunces and Manrope (Google Fonts)

3.3.3 Hardware Interface

Users need only a computer or mobile phone with a web browser. For hosting during development, any computer able to run MAMP is sufficient – a dual-core processor, 4 GB RAM and about 500 MB of free disk space.

3.3.4 Communication Interface

The browser communicates with the Apache server using HTTP/HTTPS. PHP connects to the MySQL database through PDO over TCP (127.0.0.1, port 8889 on MAMP). Sessions are maintained with a secure session cookie.

3.4 System Features

The main functional features of the system are summarised in the table below.

Table 2: System features

Id	Feature	Description
F1	Public pages	Eleven responsive pages with data read from the database.
F2	Contact enquiry	Validated form; each record is saved with status “new”.
F3	Admin login / logout	Password verification; logout destroys the session and redirects to index.php.
F4	Course management	Add, edit, delete, activate / deactivate, search, filter and image upload.
F5	Category management	Categories and subcategories; deletion is blocked if courses exist.
F6	Tutor management	CRUD with photo upload; deletion asks to reassign courses first.
F7	Admin members	Roles, custom permissions, active / inactive, password reset, delete.
F8	Messages & photos	View, mark read, archive and delete enquiries; replace site photos.
F9	Activity log	Records login, logout and every create / update / delete action.
F10	Dashboard	Counts, category chart, latest enquiries and recent activity.

3.5 Other Non-functional Requirements

3.5.1 Performance Requirement

Pages should load within two to three seconds on a normal connection. Animations use only CSS transform and opacity so that scrolling remains smooth, and database queries use indexes and prepared statements so that results are returned quickly.

3.5.2 Security Requirement

Security was given high priority because the Admin Portal controls the whole website:



- Passwords are stored as bcrypt hashes using password_hash() and checked with password_verify()
- Every form carries a CSRF token, and the session ID is regenerated at login
- All SQL queries use prepared statements to prevent SQL injection
- All output is escaped with htmlspecialchars() to prevent XSS
- Permissions are checked on the server for every page and every request
- Uploads are checked by real image content, size-limited and saved with random names
- Deactivated admins are signed out automatically on their next request

3.5.3 Safety Requirement

Safety rules prevent accidental loss of data: a confirmation dialog appears before every delete or deactivate action; an admin cannot delete or deactivate his own account or the last active Super Admin; a tutor who teaches courses cannot be deleted until those courses are reassigned; and regular database backups from phpMyAdmin are recommended.

3.5.4 Software Quality Attributes

- **Usability** – consistent layout, clear icons and helpful messages
- **Reliability** – the public site falls back to built-in data if the database is empty
- **Maintainability** – shared helper functions in includes/config.php and includes/data.php
- **Portability** – runs on any server that supports PHP and MySQL

3.6 Feasibility Study

A feasibility study was carried out to confirm that the project could be completed successfully with the available time, skills and resources.

3.6.1 Operational Feasibility

The Admin Portal uses simple tables, buttons and forms, so the academy administrator can use it with very little training. Role-based access allows work to be shared safely among staff. The system therefore fits easily into the academy’s daily work and is operationally feasible.

3.6.2 Technical Feasibility

PHP, MySQL, HTML, CSS and JavaScript are mature, well-documented and supported by almost every hosting provider. MAMP provides the complete server environment on a personal



computer, and the team already had the necessary skills. The project is therefore technically feasible.

3.6.3 Economic Feasibility

All the tools used – PHP, MySQL, Apache, MAMP (free edition), a code editor and Google Fonts – are free of cost. The only future expense is a domain name and web hosting, which is low. Because the administrator can update content without a developer, there is no recurring maintenance cost. The project is economically feasible.

3.6.4 Schedule Feasibility

The work was divided into clear phases – requirements, design, public website, database, admin portal, roles, testing and documentation – and completed between August and October 2026, as shown in the Gantt chart in Section 4.5. The project is therefore feasible within the academic schedule.

CHAPTER 4

SYSTEM ANALYSIS AND MODELING

4.1 Use Case Diagram

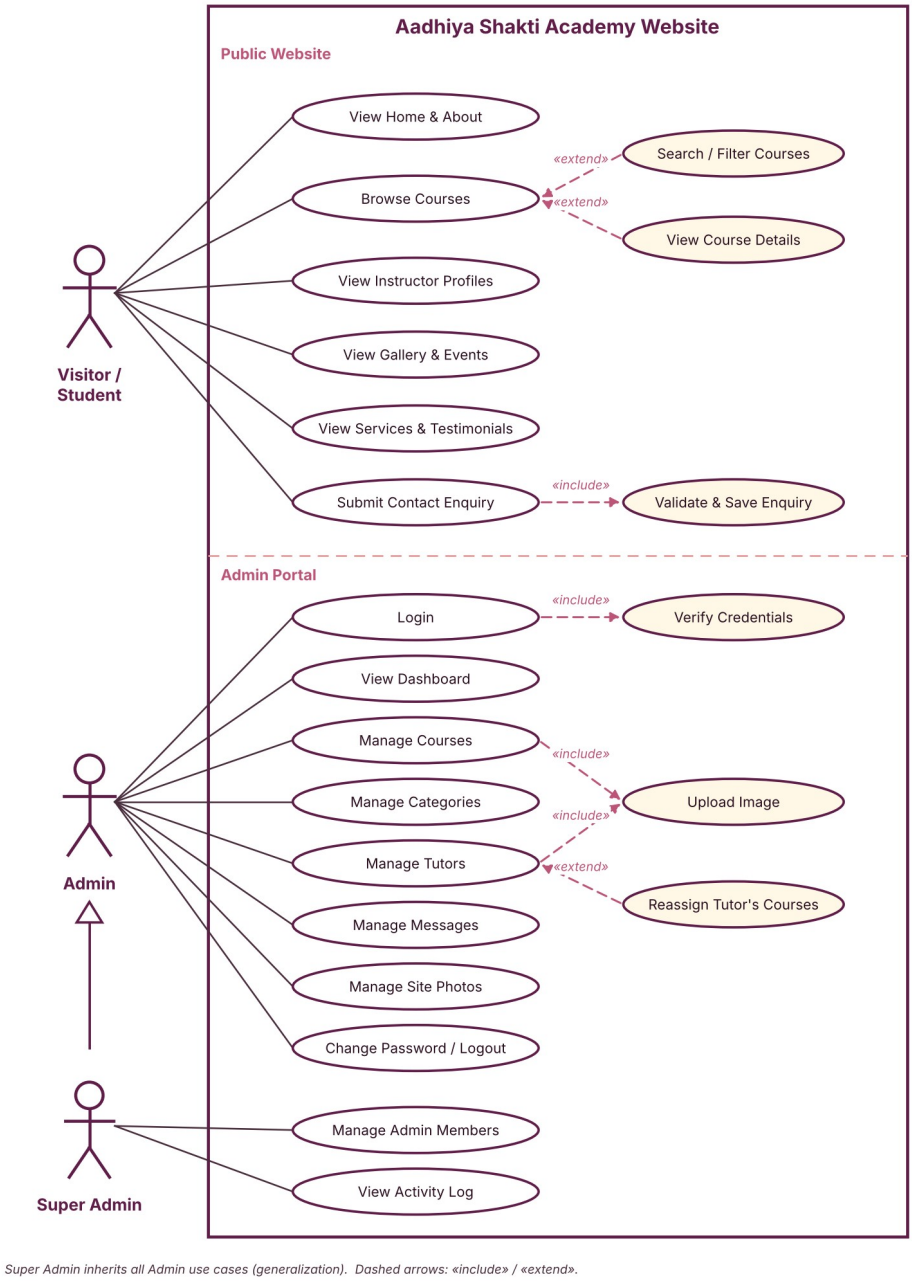


Figure 2: Use case diagram – Visitor, Admin and Super Admin

The diagram uses standard UML notation: stick figures for actors, ovals for use cases, solid lines for associations and dashed arrows for «include» and «extend». The Visitor uses only the public



pages and the contact form; submitting an enquiry always includes validating and saving it. The Admin manages the website after logging in (which includes verifying credentials); managing courses and tutors includes uploading images, and deleting a tutor may be extended by reassigning that tutor's courses. The Super Admin inherits all Admin use cases and can also manage admin members and view the activity log. Other admin roles (Content, Course and Tutor Manager) use the same Admin use cases limited by their permissions.

Role – Permission Matrix

Table 3: Default permissions of each role (can be customised per admin)

Permission	Super Admin	Admin	Content Mgr	Course Mgr	Tutor Mgr
manage_courses	✔	✔	–	✔	–
manage_course_images	✔	✔	✔	✔	–
manage_tutors	✔	✔	–	–	✔
manage_tutor_images	✔	✔	✔	–	✔
manage_categories	✔	✔	–	✔	–
manage_photos	✔	✔	✔	–	–
manage_messages	✔	✔	✔	–	–
manage_admins	✔	–	–	–	–

4.2 Normalization and E-R Diagram

Normalization is the process of organizing the columns (attributes) and tables (relations) of a relational database to reduce data redundancy and improve data integrity. The steps below show how the Aadhiya Shakti Academy data was normalized, using sample records from the project database.

(A) De-normalized Form

Before normalization, all course information was kept in one large record. Category, subcategory and tutor details are repeated for every course, and the "outcomes" column holds many values in one cell.

Course ID	Course Title	Level	Price	Category	Subcategory
1	Classical Dance Foundation	Beginner	₹2,499/month	Dance	Classical Dance
2	Kathak Foundations	Beginner	₹2,499/month	Dance	Classical Dance
3	Folk Dance Essentials	Beginner	₹1,999/month	Dance	Folk Dance



Tutor Name	Specialization	Tutor Category	Outcomes
Dr. Kavya Nair	Classical Dance (Bharatanatyam)	Dance	Correct posture; Hand gestures; Rhythm counting; First performance
Meera Iyer	Classical Dance (Kathak)	Dance	Basic tatkar; Controlled chakkars; Hand gestures; Thumri piece
Ananya Rao	Folk & Garba Dance	Dance	Folk footwork; Group formation; Prop work; Group performance

(B) First Normal Form (1NF)

In 1NF every cell holds a single (atomic) value and every row is identified by a primary key. The repeating “outcomes” values are stored one per line, and the data is divided into separate tables.

1) Table Name: categories

Primary Key: id (Auto Increment)

id	slug	name	image_key
1	dance	Dance	category_dance
2	singing	Singing	category_singing
3	choreography	Choreography	category_choreography

2) Table Name: subcategories

Primary Key: id (Auto Increment)

id	category_id	slug	name
1	1	classical	Classical Dance
2	1	folk	Folk Dance
3	1	garba	Garba / Regional Dance
4	1	bollywood	Bollywood Dance

3) Table Name: tutors

Primary Key: id (Auto Increment)

id	name	slug	specialization	category	status
1	Royan Mehta	royan-mehta	Bollywood & Contemporary Dance	dance	active
2	Dr. Kavya Nair	dr-kavya-nair	Classical Dance (Bharatanatyam)	dance	active
3	Meera Iyer	meera-iyer	Classical Dance (Kathak)	dance	active
4	Ananya Rao	ananya-rao	Folk & Garba Dance	dance	active



4) Table Name: courses

Primary Key: id (Auto Increment)

id	category_id	subcategory_id	instructor_slug	title	level	duration	price
1	1	1	dr-kavya-nair	Classical Dance Foundation	Beginner	12 Weeks	₹2,499/month
2	1	1	meera-iyer	Kathak Foundations	Beginner	12 Weeks	₹2,499/month
3	1	2	ananya-rao	Folk Dance Essentials	Beginner	8 Weeks	₹1,999/month
4	1	3	ananya-rao	Garba & Regional Dance	All Levels	6 Weeks	₹1,799/month

5) Table Name: admin_users

Primary Key: id (Auto Increment)

id	username	role	status	permissions
1	admin	super_admin	active	NULL (all permissions)

6) Table Name: admin_activity_log

Primary Key: id (Auto Increment)

id	admin_id	admin_username	action	description	created_at
1	1	admin	login	Logged in.	2026-10-05 10:15
2	1	admin	course_update	Edited course "Kathak Foundations"	2026-10-05 10:21
3	1	admin	logout	Logged out.	2026-10-05 10:40

7) Table Name: contact_messages

Primary Key: id (Auto Increment)

id	name	email	phone	service	message	status
1	Riya Sharma	riya@example.com	98XXXXXX01	Dance Training	Batch timings for Kathak classes?	new
2	Karan Joshi	karan@example.com	98XXXXXX02	Choreography	Wedding sangeet choreography	read

8) Table Name: site_images

Primary Key: image_key

image_key	filename	updated_at
about_main	about_main-3f2a9c.jpg	2026-10-05 11:02
category_dance	category_dance-8b1e4d.jpg	2026-10-05 11:05



(C) Second Normal Form (2NF)

In 2NF every non-key attribute is fully dependent on the whole primary key. Category, subcategory and tutor details no longer repeat inside each course; they live in their own tables and are linked through foreign keys.

1) Table Name: categories

Primary Key: id (Auto Increment)

id (PK)	slug	name	image_key
1	dance	Dance	category_dance
2	singing	Singing	category_singing
3	choreography	Choreography	category_choreography

2) Table Name: subcategories

Primary Key: id (Auto Increment) **Foreign Key:** category_id

id (PK)	category_id (FK)	slug	name
1	1	classical	Classical Dance
2	1	folk	Folk Dance
3	1	garba	Garba / Regional Dance
4	1	bollywood	Bollywood Dance

3) Table Name: tutors

Primary Key: id (Auto Increment)

id (PK)	name	slug (UNIQUE)	specialization	category	status
1	Royan Mehta	royan-mehta	Bollywood & Contemporary Dance	dance	active
2	Dr. Kavya Nair	dr-kavya-nair	Classical Dance (Bharatanatyam)	dance	active
3	Meera Iyer	meera-iyer	Classical Dance (Kathak)	dance	active
4	Ananya Rao	ananya-rao	Folk & Garba Dance	dance	active

4) Table Name: courses

Primary Key: id (Auto Increment) **Foreign Key:** category_id, subcategory_id, instructor_slug

id (PK)	category_id (FK)	subcategory_id (FK)	instructor_slug (FK)	title	level	duration	price
1	1	1	dr-kavya-nair	Classical Dance Foundation	Beginner	12 Weeks	₹2,499/month
2	1	1	meera-iyer	Kathak Foundations	Beginner	12 Weeks	₹2,499/month
3	1	2	ananya-rao	Folk Dance Essentials	Beginner	8 Weeks	₹1,999/month
4	1	3	ananya-rao	Garba & Regional Dance	All Levels	6 Weeks	₹1,799/month



5) Table Name: admin_activity_log

Primary Key: id (Auto Increment) **Foreign Key:** admin_id

id (PK)	admin_id (FK)	admin_username	action	description	created_at
1	1	admin	login	Logged in.	2026-10-05 10:15
2	1	admin	course_update	Edited course "Kathak Foundations"	2026-10-05 10:21
3	1	admin	logout	Logged out.	2026-10-05 10:40

(D) Third Normal Form (3NF)

In 3NF no non-key attribute depends on another non-key attribute (no transitive dependency). The tutor's specialization depends on the tutor, not on the course, so it is stored only in **tutors**; the category name depends on the category, so it is stored only in **categories**. Admin accounts, activity log, contact enquiries and site photos are kept in separate tables. The final database therefore contains nine tables, all in 3NF.

Table	Primary Key	Foreign Key(s)
categories	id	–
subcategories	id	category_id → categories.id
courses	id	category_id → categories.id, subcategory_id → subcategories.id, instructor_slug → tutors.slug
tutors	id	–
admin_users	id	–
admin_activity_log	id	admin_id → admin_users.id
contact_messages	id	–
site_images	image_key	– (matched by categories.image_key)
course_overrides	course_slug	– (legacy, kept for compatibility)

Entities and Relationships

Relationship	Cardinality	Meaning
categories → subcategories	1 : N	A category has many subcategories
categories → courses	1 : N	A category contains many courses
subcategories → courses	1 : N	A subcategory may contain many courses
tutors → courses	1 : N	A tutor can teach many courses (instructor_slug)
admin_users → admin_activity_log	1 : N	An admin creates many log entries
categories → site_images	1 : 0..1	A category card photo can be replaced by one uploaded image (matched by image_key)

The tables **contact_messages** (visitor enquiries) and **course_overrides** (an older price-override table kept only for backward compatibility) have no foreign-key relationship with other tables, so they are not drawn in the E-R diagram. They are described in the Data Dictionary.

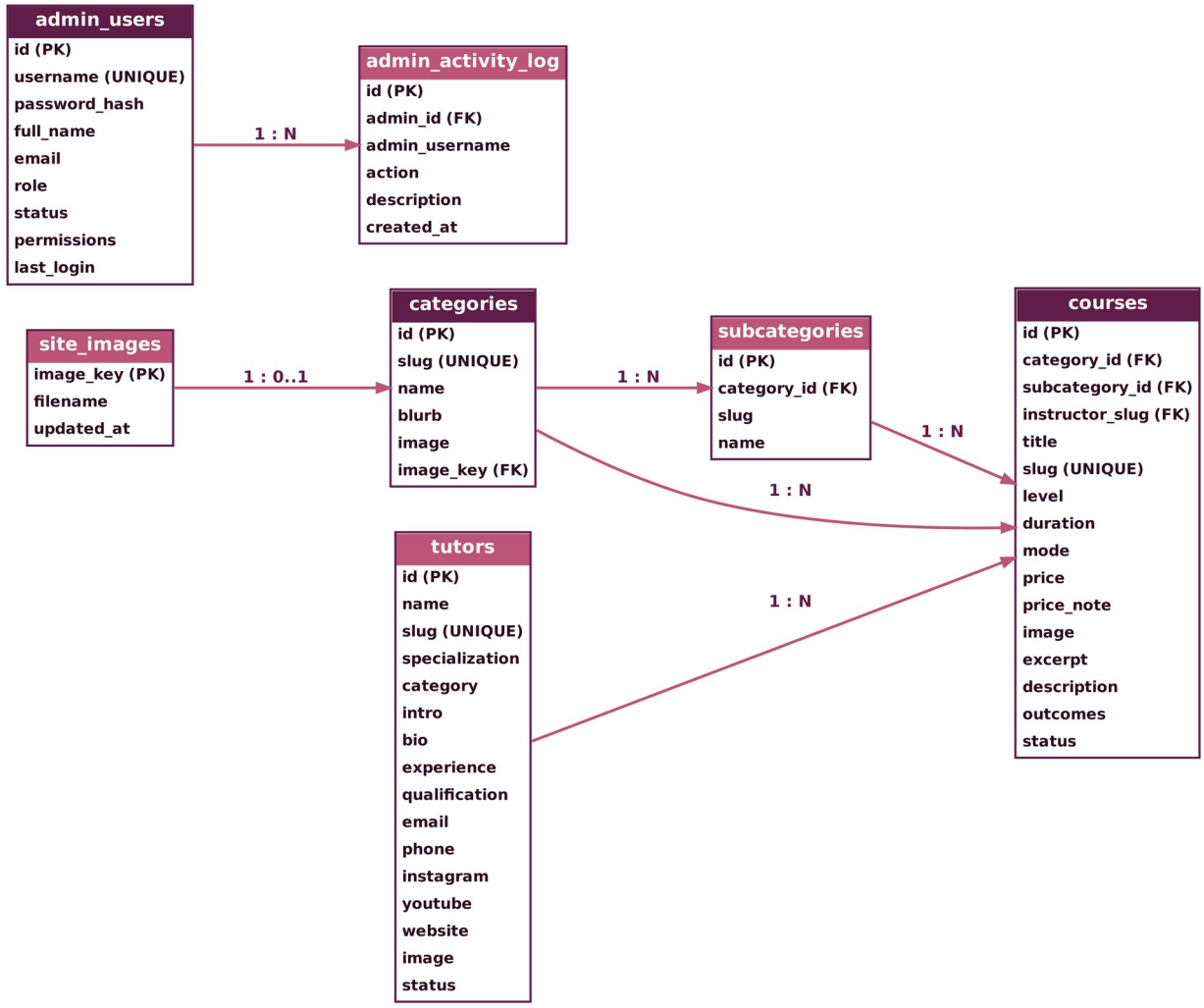


Figure 3: Entity–Relationship diagram of aadhiya_shakti_site

4.3 Data Dictionary

Table: categories

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Category id
2	slug	VARCHAR(80)	NO	Unique	URL key (dance …)
3	name	VARCHAR(120)	NO	-	Category name
4	blurb	TEXT	YES	-	Short description
5	image	VARCHAR(255)	YES	-	Image path / URL
6	image_key	VARCHAR(80)	YES	-	Key for photo override



Table: subcategories

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Subcategory id
2	category_id	INT UNSIGNED	NO	Foreign → categories.id	Parent category
3	slug	VARCHAR(80)	NO	Unique with category	URL key
4	name	VARCHAR(120)	NO	-	Subcategory name

Table: courses

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Course id
2	category_id	INT UNSIGNED	NO	Foreign → categories.id	Category
3	subcategory_id	INT UNSIGNED	YES	Foreign → subcategories.id	Subcategory (SET NULL on delete)
4	instructor_slug	VARCHAR(120)	YES	tutors.slug	Assigned tutor
5	title	VARCHAR(190)	NO	-	Course title
6	slug	VARCHAR(190)	NO	Unique	URL key
7	level	VARCHAR(60)	NO	-	Beginner / Intermediate …
8	duration	VARCHAR(80)	NO	-	e.g. 12 Weeks
9	mode	VARCHAR(60)	NO	-	Online / In-Studio …
10	price	VARCHAR(80)	NO	-	Displayed price (no payment)
11	price_note	VARCHAR(40)	NO	-	Sample Pricing / Starting From
12	image	VARCHAR(255)	YES	-	Image path or URL
13	excerpt	VARCHAR(400)	YES	-	Short text
14	description	TEXT	YES	-	Long text
15	outcomes	TEXT	YES	-	One outcome per line
16	status	ENUM	NO	-	active / inactive
17	created_at, updated_at	TIMESTAMP	NO	-	Auto timestamps

Table: tutors

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Tutor id
2	name	VARCHAR(120)	NO	-	Tutor name
3	slug	VARCHAR(120)	NO	Unique	URL key
4	specialization	VARCHAR(190)	NO	-	Specialty



Sr.	Column	Data Type	Null	Key / Links	Comments
5	category	VARCHAR(80)	NO	-	dance / singing / choreography
6	intro	VARCHAR(400)	YES	-	Short intro
7	bio	TEXT	YES	-	Biography
8	experience, qualification	VARCHAR	YES	-	Profile details
9	email, phone	VARCHAR	YES	-	Contact details
10	instagram, youtube, website	VARCHAR(190)	YES	-	Social links
11	image	VARCHAR(255)	YES	-	Photo path
12	status	ENUM	NO	-	active / inactive

Table: admin_users

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Admin id
2	username	VARCHAR(60)	NO	Unique	Login name
3	password_hash	VARCHAR(255)	NO	-	bcrypt hash
4	full_name	VARCHAR(120)	YES	-	Full name
5	email	VARCHAR(190)	YES	-	E-mail
6	role	VARCHAR(30)	NO	-	super_admin, admin, content_manager …
7	status	ENUM	NO	-	active / inactive
8	permissions	TEXT	YES	-	JSON list of permissions
9	created_at, last_login	TIMESTAMP	-	-	Time stamps

Table: admin_activity_log

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Log id
2	admin_id	INT UNSIGNED	YES	admin_users.id	Who
3	admin_username	VARCHAR(60)	NO	-	Username copy
4	action	VARCHAR(60)	NO	-	login, course_update …
5	description	VARCHAR(500)	NO	-	Readable text
6	created_at	TIMESTAMP	NO	Indexed	When

Table: contact_messages

Primary Key: id

Sr.	Column	Data Type	Null	Key / Links	Comments
1	id	INT UNSIGNED	NO	Primary, Auto	Enquiry id
2	name	VARCHAR(120)	NO	-	Visitor name

Sr.	Column	Data Type	Null	Key / Links	Comments
3	email	VARCHAR(190)	NO	-	Visitor e-mail
4	phone	VARCHAR(30)	YES	-	Phone
5	service	VARCHAR(60)	NO	-	Selected service
6	message	TEXT	NO	-	Enquiry text
7	status	ENUM	NO	Indexed	new / read / archived
8	created_at	TIMESTAMP	NO	Indexed	Received time

Table: site_images

Primary Key: image_key

Sr.	Column	Data Type	Null	Key / Links	Comments
1	image_key	VARCHAR(60)	NO	Primary	Spot name (about_main …)
2	filename	VARCHAR(255)	NO	-	Uploaded file name
3	updated_at	TIMESTAMP	NO	-	Last change

Table: course_overrides

Primary Key: course_slug

Sr.	Column	Data Type	Null	Key / Links	Comments
1	course_slug	VARCHAR(180)	NO	Primary	Course key
2	price	VARCHAR(60)	NO	-	Overriding price
3	price_note	VARCHAR(30)	NO	-	Overriding note
4	updated_at	TIMESTAMP	NO	-	Last change

4.4 Functional and Behavior Modeling

4.4.1 Context Diagram (0-Level DFD)

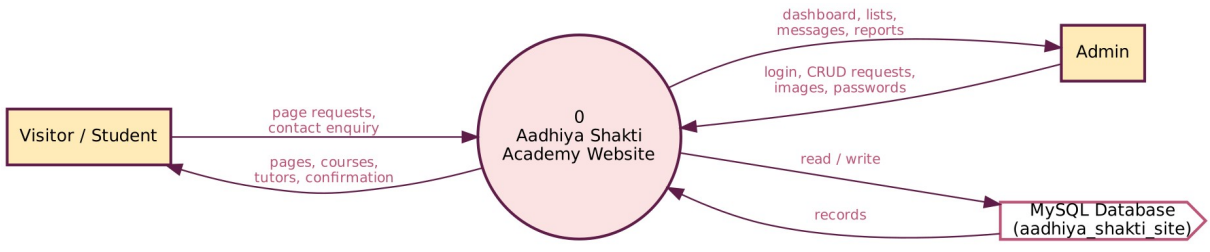


Figure 4: Context diagram

4.4.2 First Level DFD – Visitor

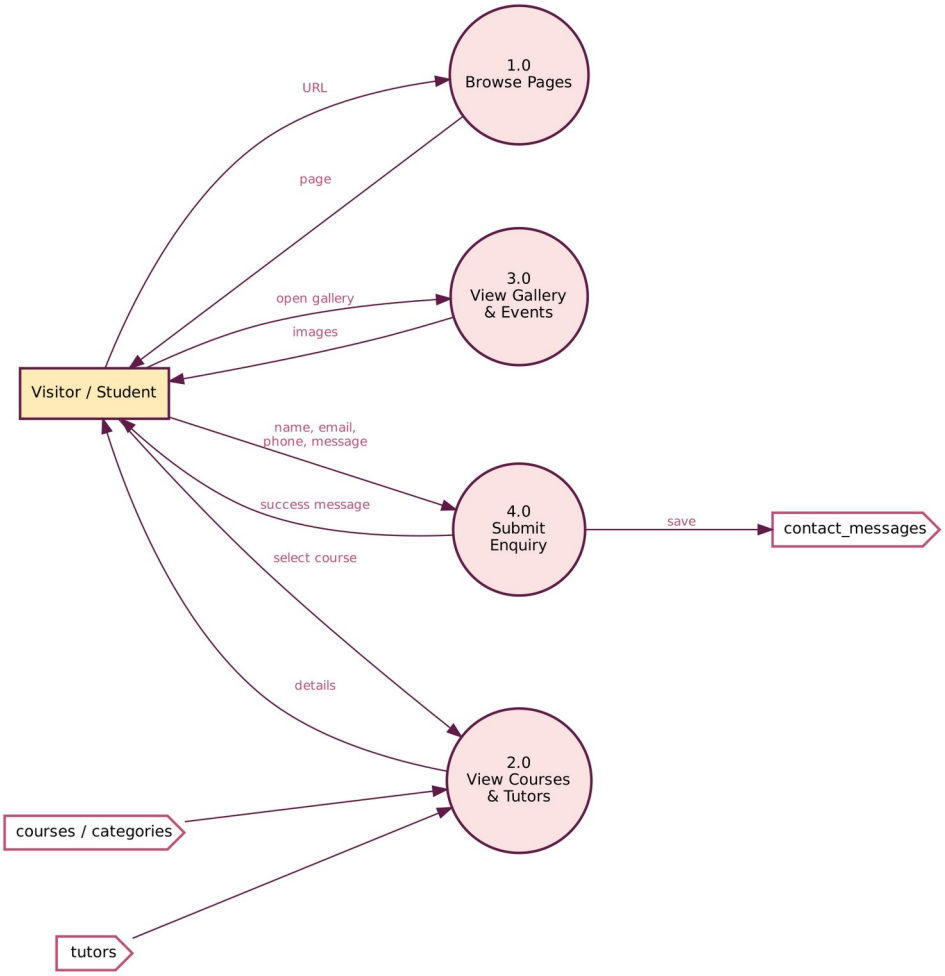


Figure 5: Level-1 DFD for the public visitor

4.4.3 First Level DFD – Admin

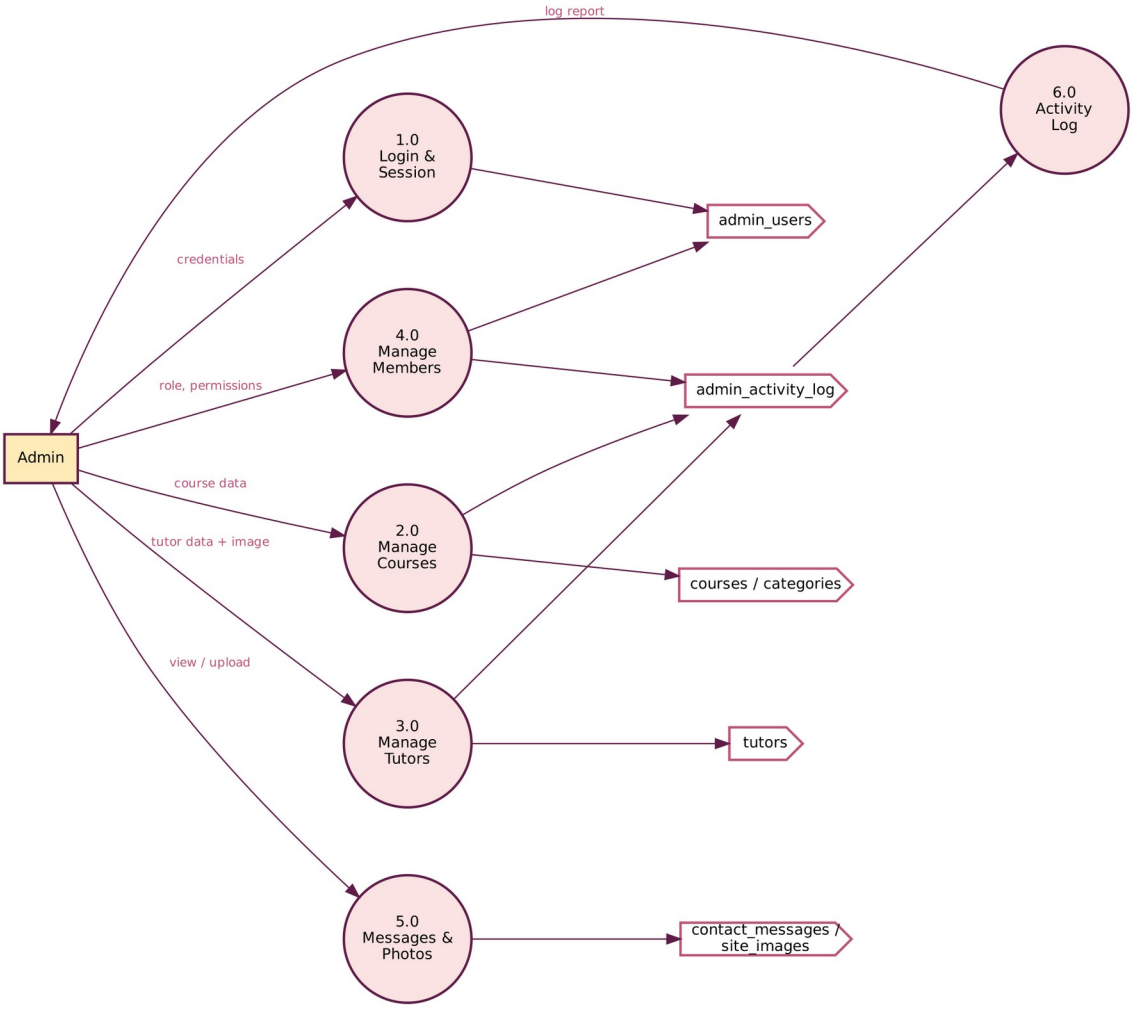


Figure 6: Level-1 DFD for the administrator

4.5 Gantt Chart

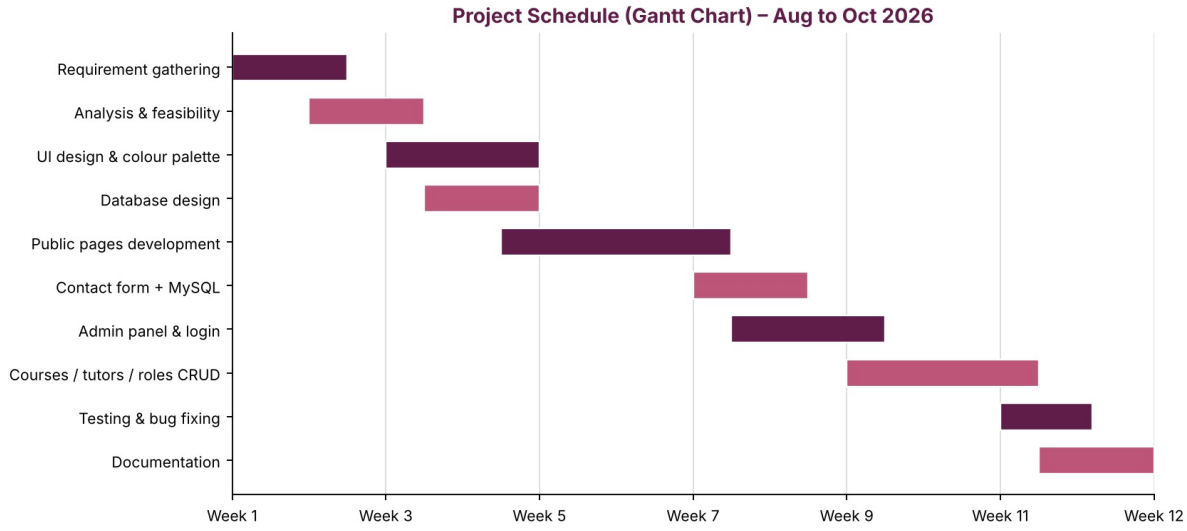


Figure 7: Project schedule (Gantt chart)



CHAPTER 5

TEST CASES

Each test case lists the steps, test data, expected result and the actual result. Results were obtained by running the steps against the working system (MySQL on port 8889). Tests of the admin permissions were repeated with direct HTTP requests to prove that the server – not only the hidden menu – enforces the rules.

Test Case No. 1: Admin Login

DESCRIPTION: Verify that only a valid admin can log in.

PRECONDITION: Admin account exists and is active.

Step	Test step	Test data	Expected result	Actual result	Status
1	Open admin/login.php	URL	Login form shown	Same as expected	PASS
2	Enter wrong password	admin / wrong	Error “Incorrect username or password”	Same as expected	PASS
3	Submit without CSRF token	valid credentials	Request rejected (session expired)	Same as expected	PASS
4	Enter correct credentials	admin / ChangeMe123!	Redirect to dashboard	Same as expected	PASS
5	Open admin/index.php without login	URL	Redirect to login.php	Same as expected	PASS

Test Case No. 2: Admin Logout

DESCRIPTION: Verify that logout destroys the session and goes to the public home page.

PRECONDITION: Admin is logged in.

Step	Test step	Test data	Expected result	Actual result	Status
1	Click Logout	Button	Redirect Location: ../index.php	Same as expected	PASS
2	Open admin/courses.php	URL	Redirect to login.php	Same as expected	PASS
3	Check activity log	Log page	“Logged out” entry recorded	Same as expected	PASS

Test Case No. 3: Contact Enquiry

DESCRIPTION: Verify that the contact form validates and stores enquiries.

PRECONDITION: Database is imported.

Step	Test step	Test data	Expected result	Actual result	Status
1	Submit empty / invalid form	Blank message, bad e-mail	Not stored, error shown	Same as expected	PASS



Step	Test step	Test data	Expected result	Actual result	Status
2	Submit valid form	Name, e-mail, phone, service, message	Record saved in contact_messages	Same as expected	PASS
3	Open admin Messages	Admin login	Enquiry visible with status “new”	Same as expected	PASS

Test Case No. 4: Course Management

DESCRIPTION: Verify add, edit, search and validation of courses.

PRECONDITION: Admin has manage_courses permission.

Step	Test step	Test data	Expected result	Actual result	Status
1	Add course with image	Title, category, price, PNG file	Course saved; visible on public Courses page	Same as expected	PASS
2	Add empty required fields	Blank title / price	Error: required fields	Same as expected	PASS
3	Duplicate slug	kathak-foundations	Error: slug already used	Same as expected	PASS
4	Search courses	q = kathak	Only matching rows shown	Same as expected	PASS
5	Deactivate course	Button	Hidden from public page	Same as expected	PASS

Test Case No. 5: Tutor Management and Image Upload

DESCRIPTION: Verify tutor CRUD and secure image upload.

PRECONDITION: Admin has manage_tutors permission.

Step	Test step	Test data	Expected result	Actual result	Status
1	Add tutor with photo	Name, specialization, PNG	Saved; shown on Instructors page with uploaded photo	Same as expected	PASS
2	Upload fake image	PHP text renamed .png	Rejected: not a valid image	Same as expected	PASS
3	Deactivate tutor	Royan Mehta	Hidden on public Instructors page	Same as expected	PASS
4	Re-activate tutor	Button	Visible again	Same as expected	PASS

Test Case No. 6: Safe Tutor Deletion

DESCRIPTION: Verify that deleting a tutor with courses requires reassignment.

PRECONDITION: Tutor teaches at least one course.

Step	Test step	Test data	Expected result	Actual result	Status
1	Click Delete on tutor with courses	Dr. Kavya Nair	Redirect to reassign screen	Same as expected	PASS
2	Check database	tutors table	Tutor still present	Same as expected	PASS



Step	Test step	Test data	Expected result	Actual result	Status
3	Reassign and delete	royan-mehta	Tutor deleted; courses moved to selected tutor	Same as expected	PASS

Test Case No. 7: Role-based Permissions

DESCRIPTION: Verify the server blocks actions the role does not have.

PRECONDITION: Course Manager and Tutor Manager accounts created.

Step	Test step	Test data	Expected result	Actual result	Status
1	Course Manager opens tutors page	URL	HTTP 403 Access denied	Same as expected	PASS
2	Course Manager opens members page	URL	HTTP 403 Access denied	Same as expected	PASS
3	Course Manager sends direct POST to delete tutor	Forged request with valid CSRF	HTTP 403	Same as expected	PASS
4	Tutor Manager uploads tutor image without image permission	PNG file	Error: no permission to upload tutor images	Same as expected	PASS
5	Course Manager opens courses page	URL	Page opens (HTTP 200)	Same as expected	PASS

Test Case No. 8: Admin Members

DESCRIPTION: Verify account status and self-protection rules.

PRECONDITION: Super Admin is logged in.

Step	Test step	Test data	Expected result	Actual result	Status
1	Deactivate a member	Toggle	Status inactive	Same as expected	PASS
2	Open page with that member's open session	URL	Redirect to login	Same as expected	PASS
3	Inactive member logs in	Correct password	Login refused	Same as expected	PASS
4	Delete own account / last Super Admin	Delete	Refused with message	Same as expected	PASS

Test Case No. 9: Public Pages

DESCRIPTION: Verify that every public page loads without errors.

PRECONDITION: Site running on Apache or PHP server.

Step	Test step	Test data	Expected result	Actual result	Status
1	Open index, courses, instructors, gallery	URLs	HTTP 200, no PHP warning	Same as expected	PASS
2	Open about, contact, events, services, testimonials	URLs	HTTP 200, no PHP warning	Same as expected	PASS



Step	Test step	Test data	Expected result	Actual result	Status
3	Open course-detail and instructor-detail	slug parameters	Details displayed	Same as expected	PASS

Test Case No. 10: Change Password

DESCRIPTION: Verify that a wrong current password is rejected.

PRECONDITION: Admin is logged in.

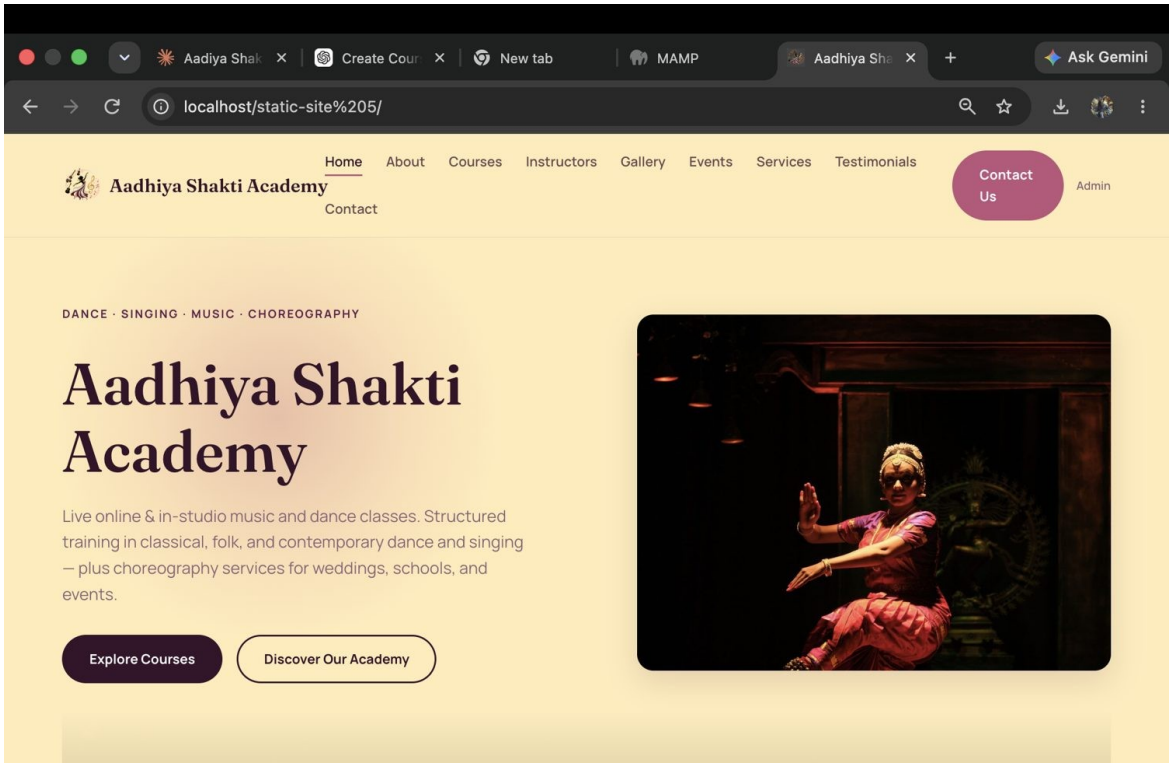
Step	Test step	Test data	Expected result	Actual result	Status
1	Enter wrong current password	bad / NewPass123!	Message: current password incorrect	Same as expected	PASS

CHAPTER 6

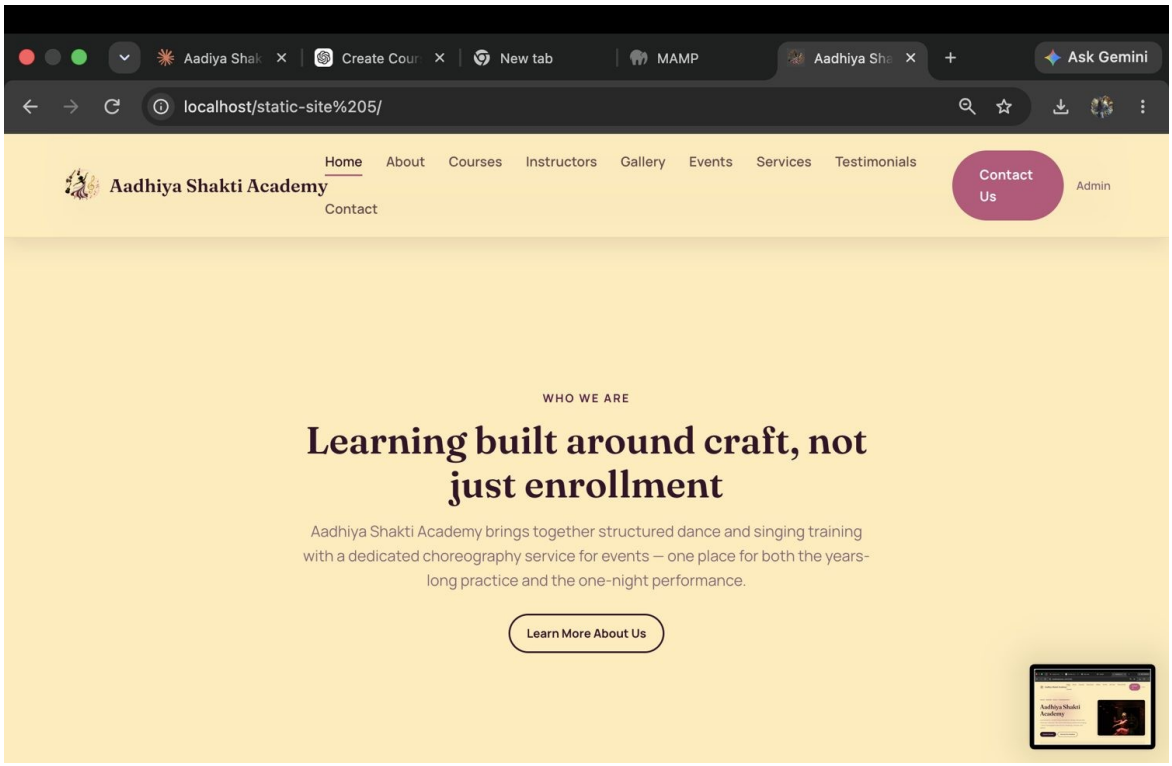
SCREEN SHOTS

This chapter shows the working screens of the public website (6.1 – 6.10) and of the Admin Portal (6.11 onwards). In the new admin screenshots, tutor and course photographs that are hosted on the internet appear as plain plum placeholders because the screenshots were taken without internet access; on a connected computer the real photographs are shown.

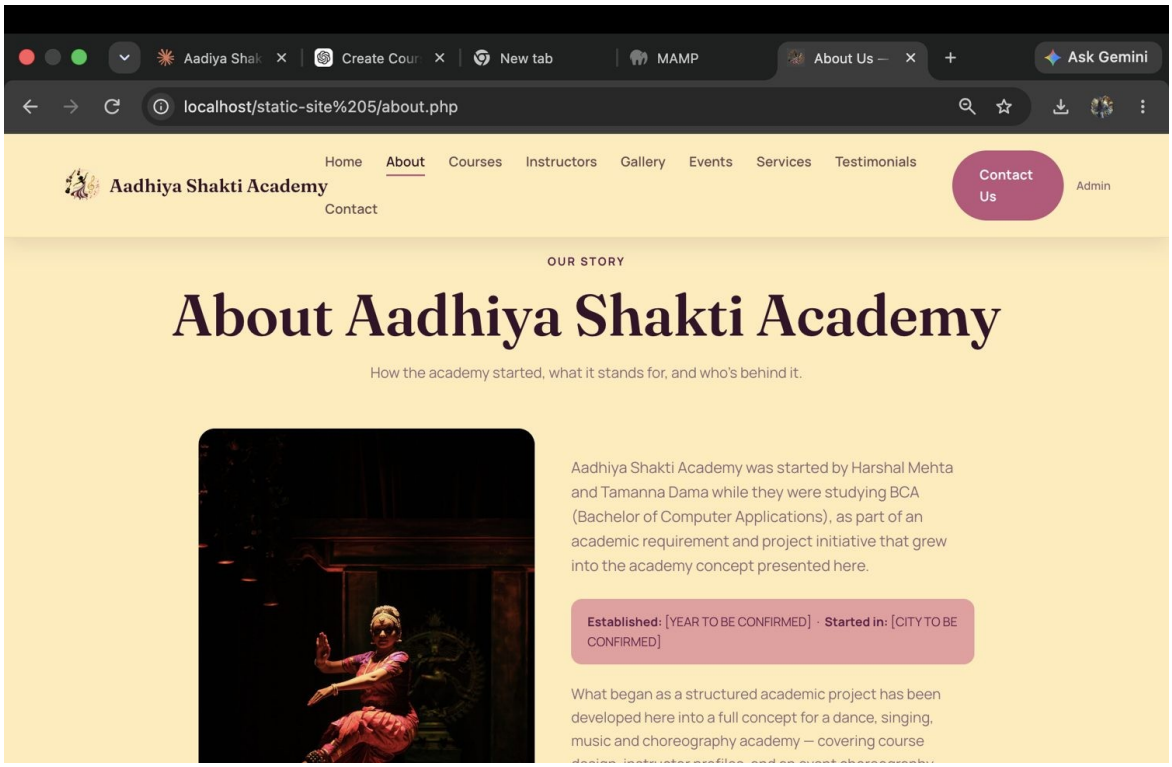
6.1 This page shows: Home page – hero section with call-to-action buttons



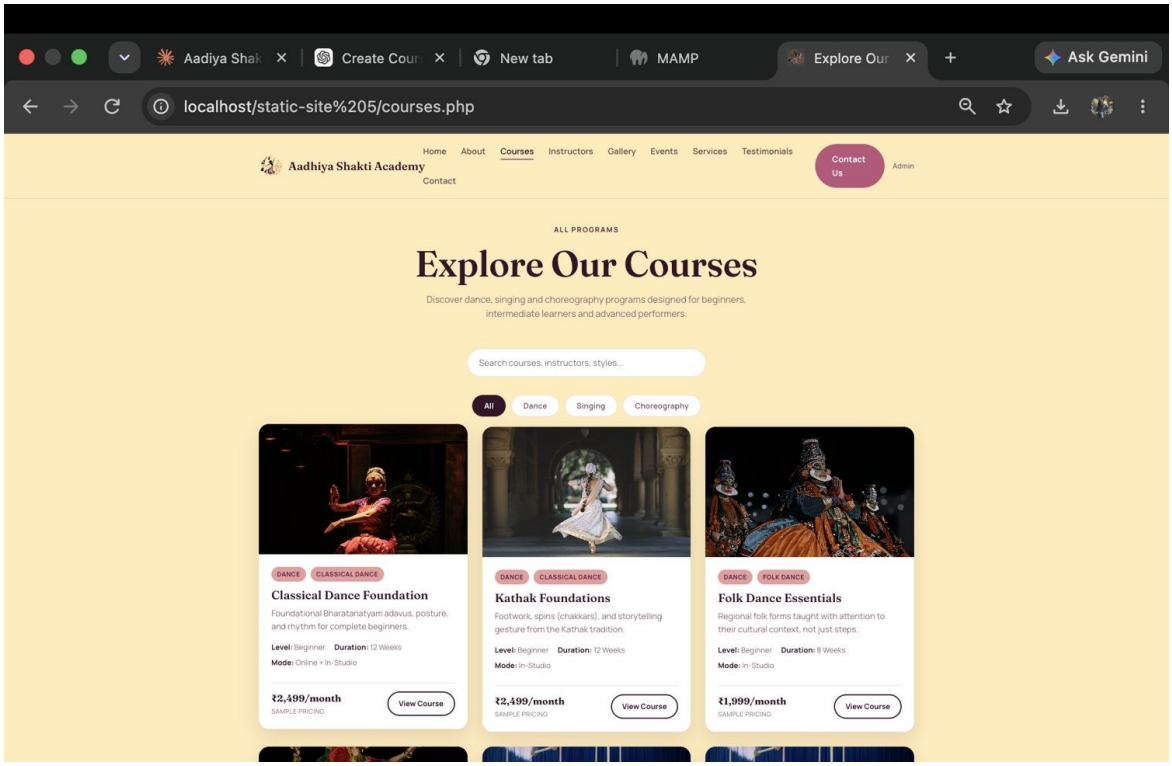
6.2 This page shows: Home page – “Learning built around craft” introduction strip



6.3 This page shows: About page – the academy story

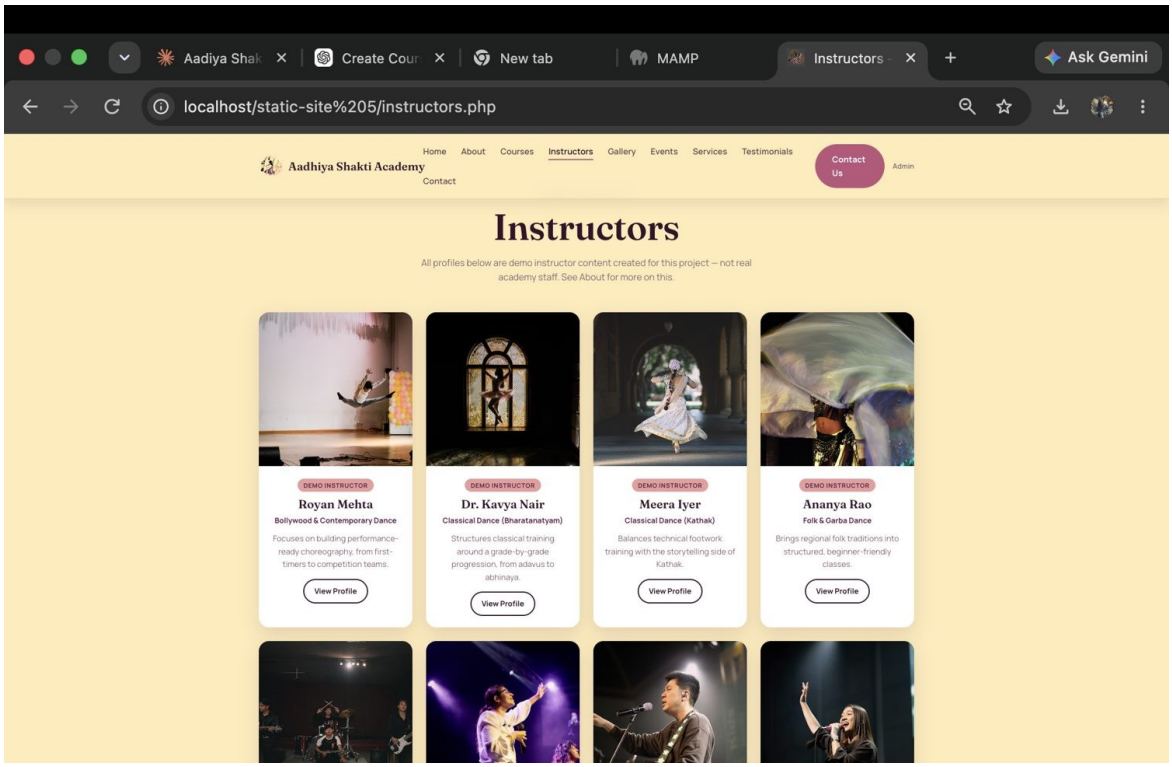


6.4 This page shows: Courses page – search box, category filter and course cards

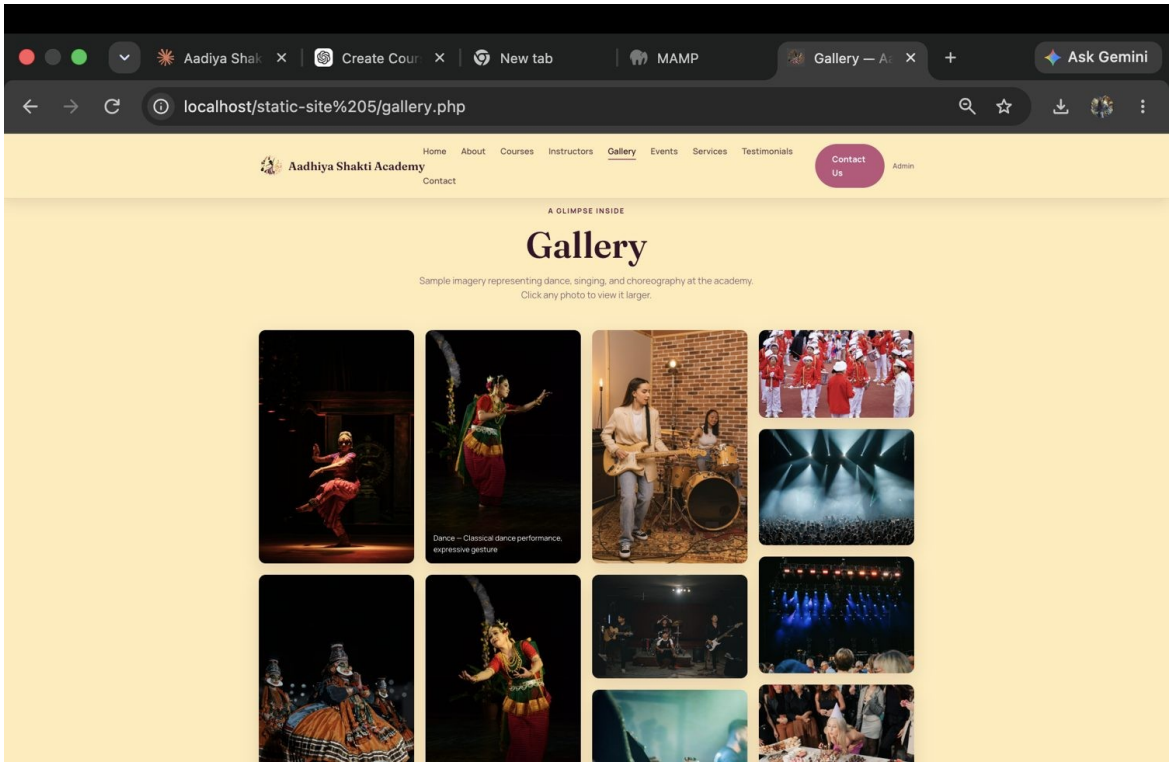




6.5 This page shows: Instructors page – tutor profile cards

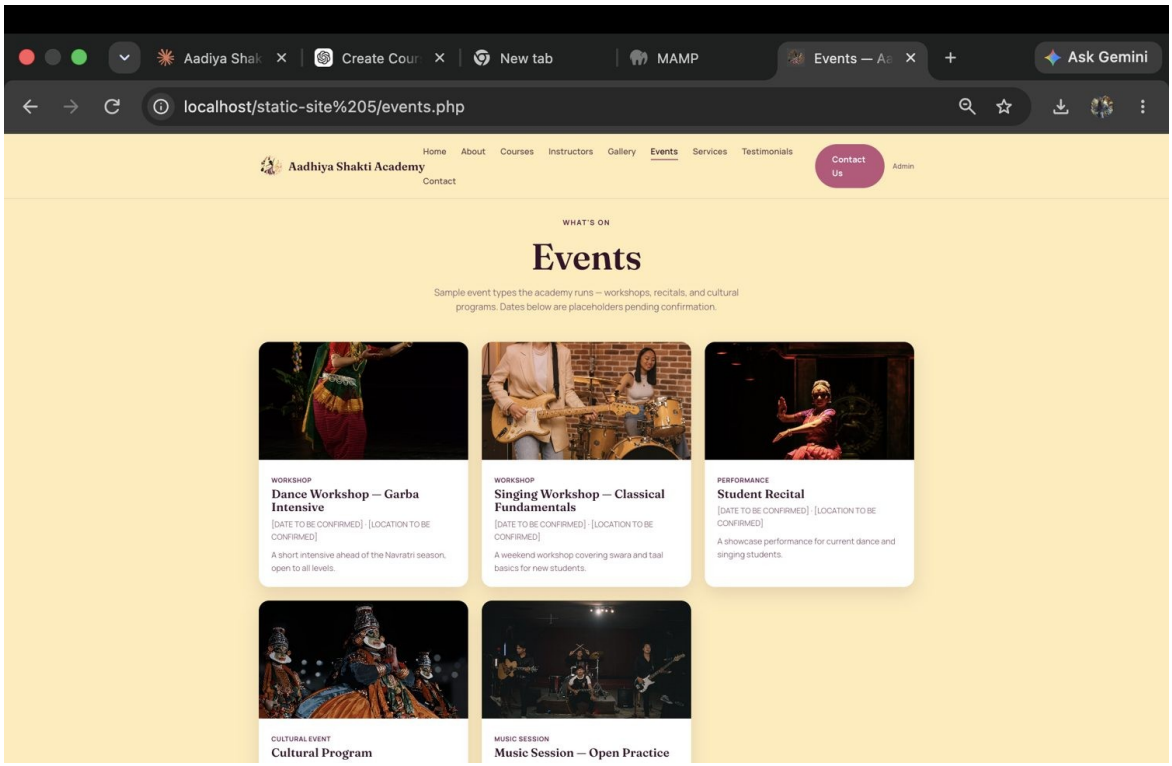


6.6 This page shows: Gallery page – masonry photo gallery

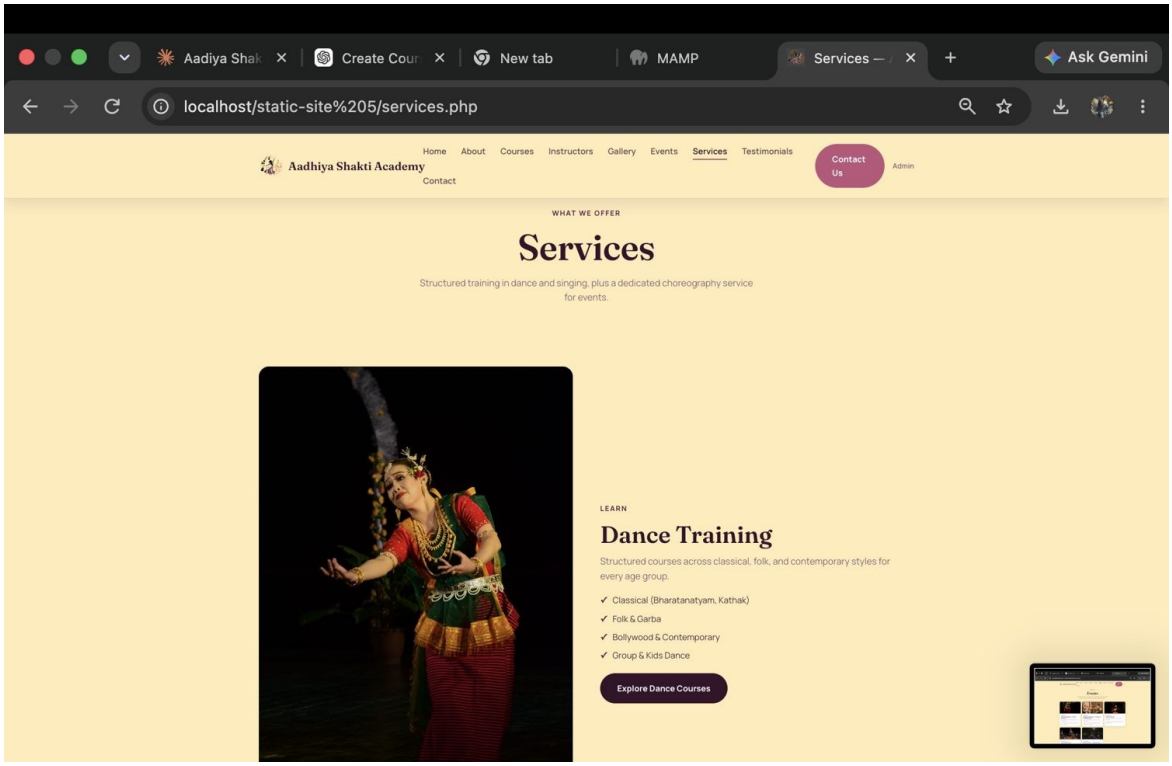




6.7 This page shows: Events page – workshops and performances

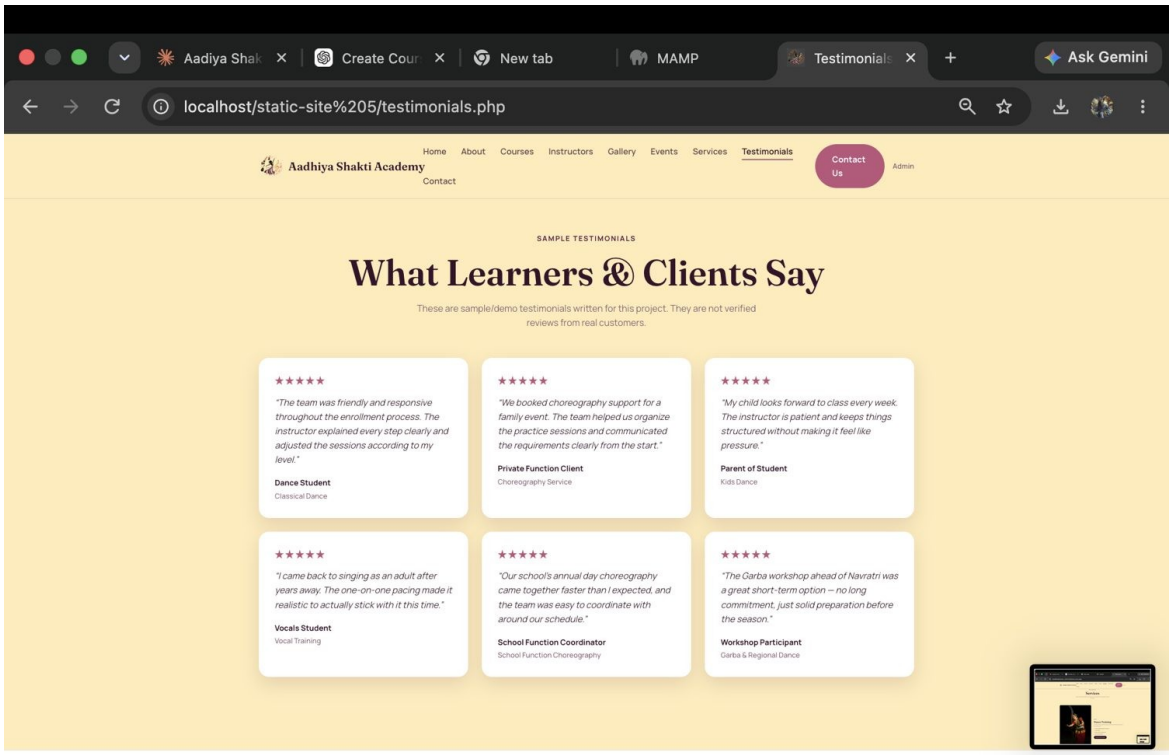


6.8 This page shows: Services page – dance training and choreography services

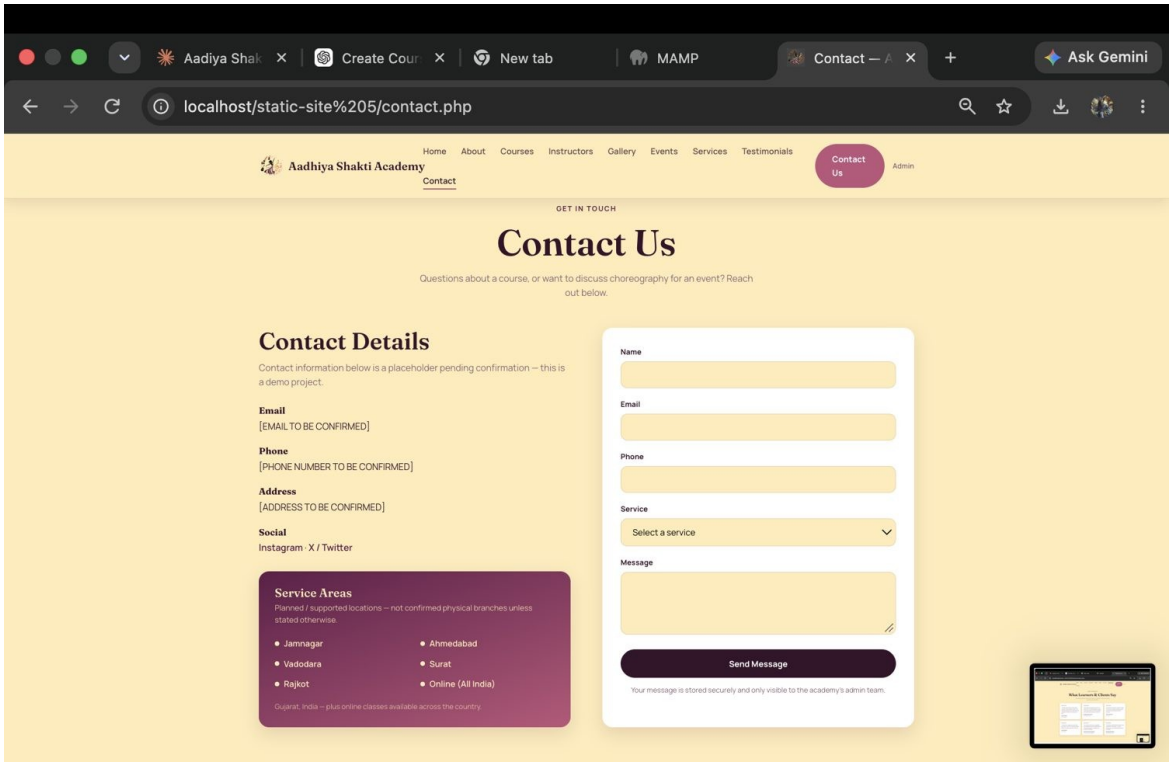




6.9 This page shows: Testimonials page – sample learner and client feedback

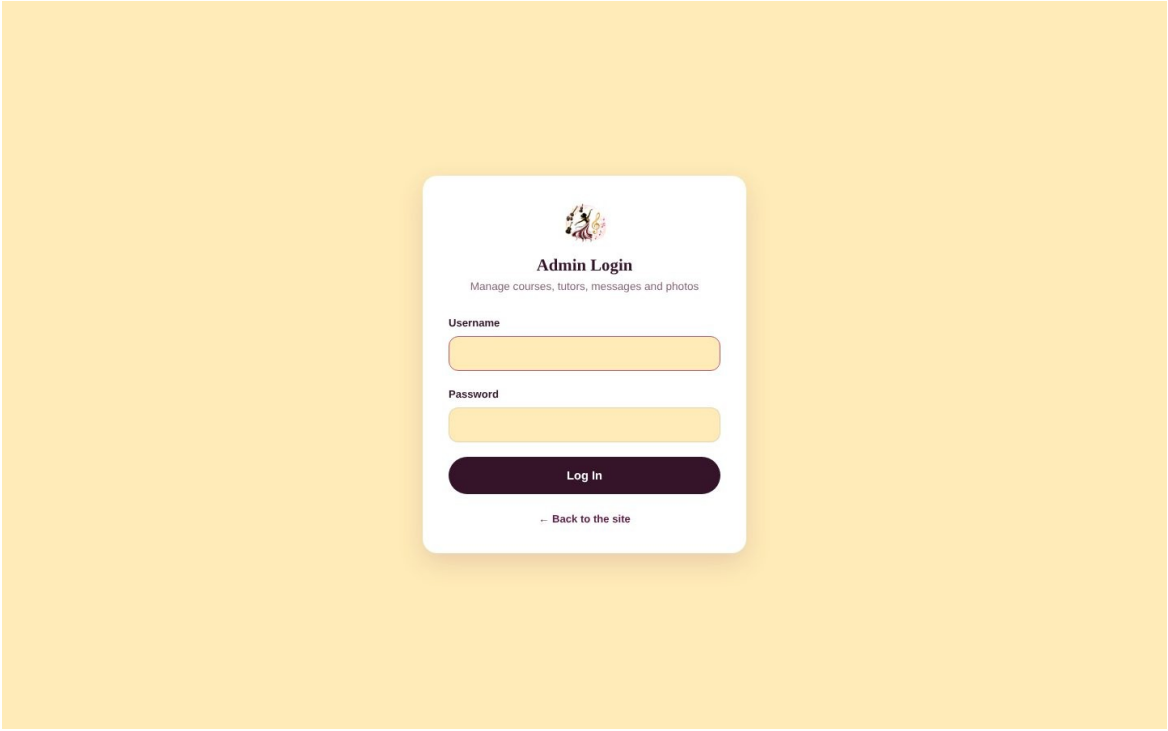


6.10 This page shows: Contact page – contact details, Service Areas panel and enquiry form

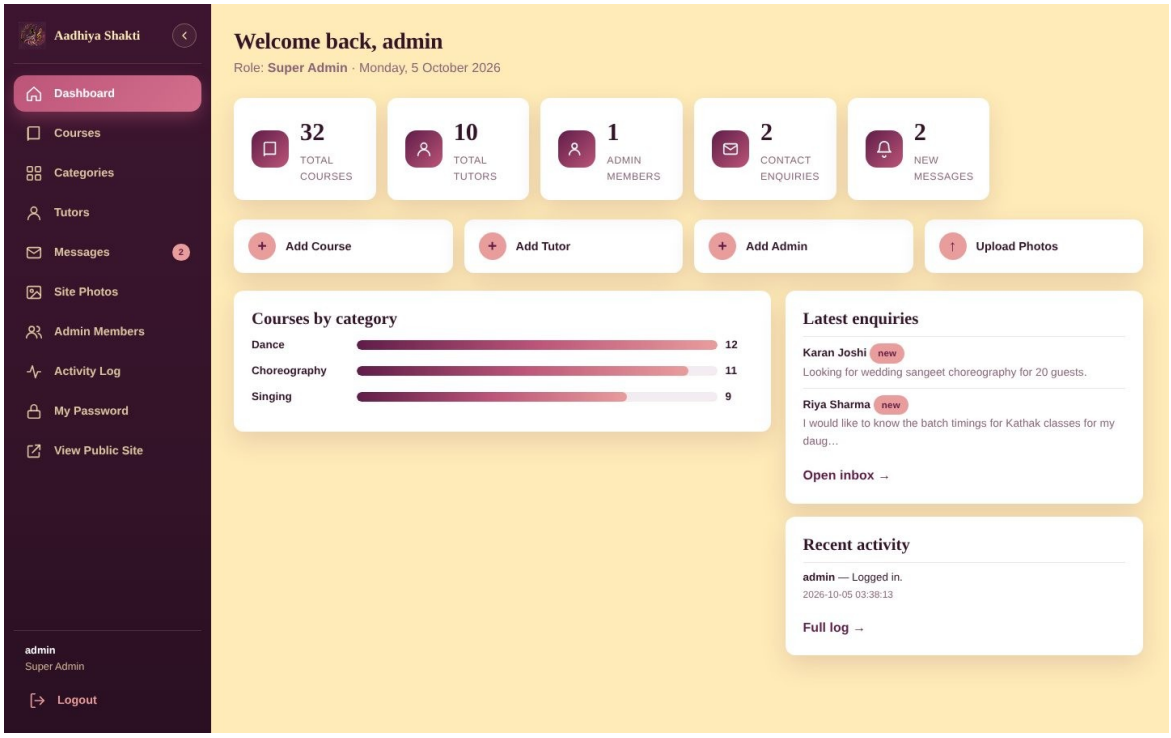




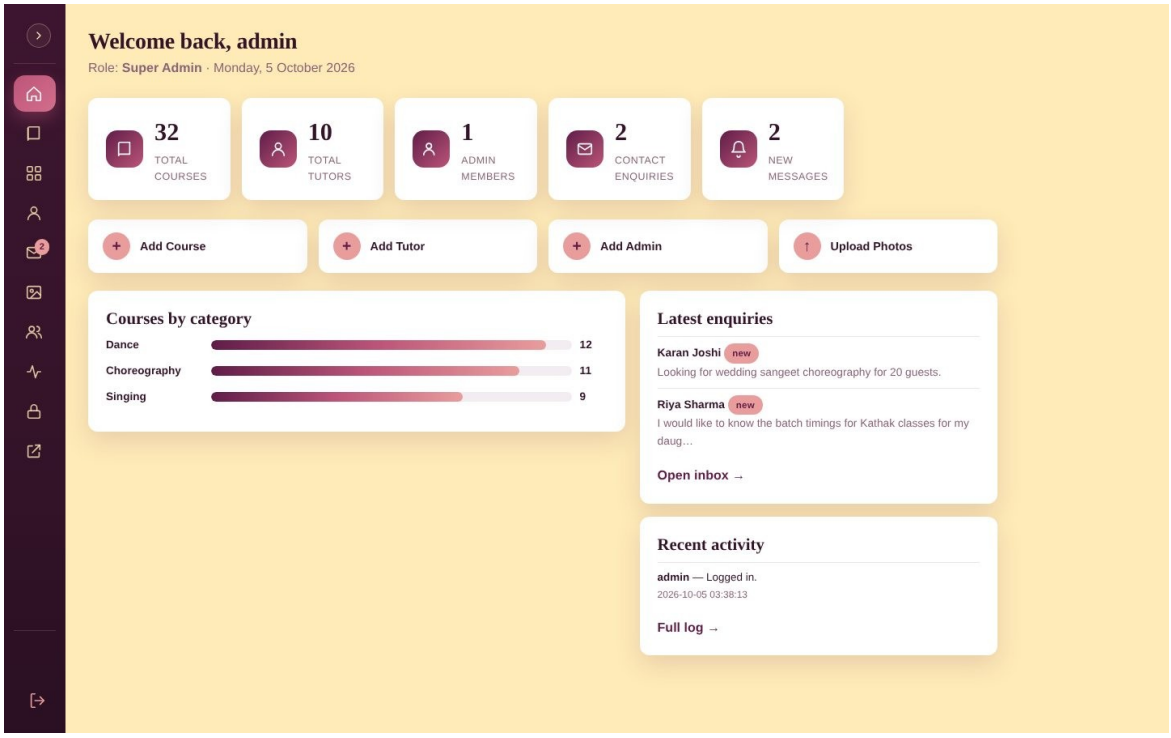
6.11 This page shows: Admin login page



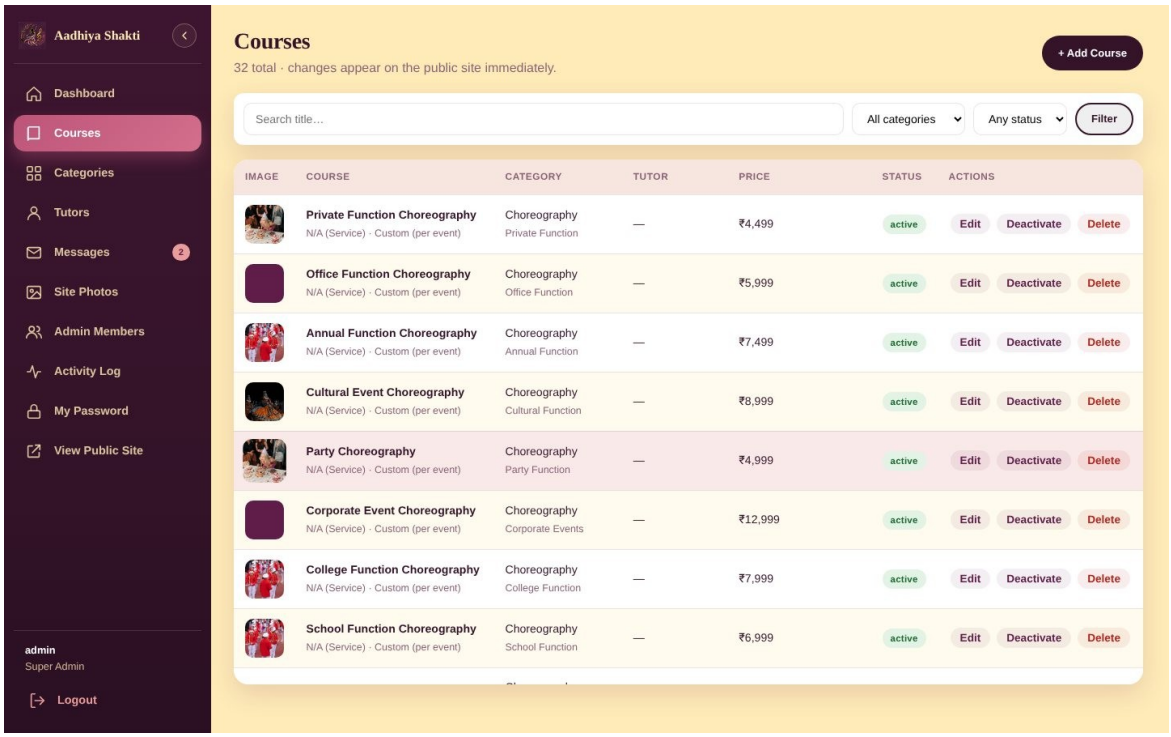
6.12 This page shows: Dashboard – statistic cards, quick actions, category bars, latest enquiries and recent activity



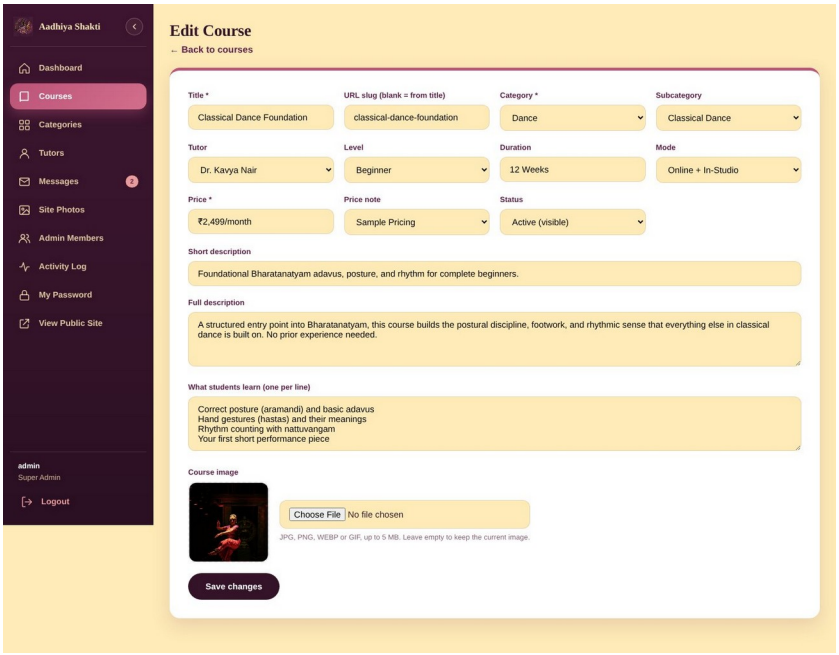
6.13 This page shows: Collapsed sidebar – icon-only mode that expands when the mouse moves over it



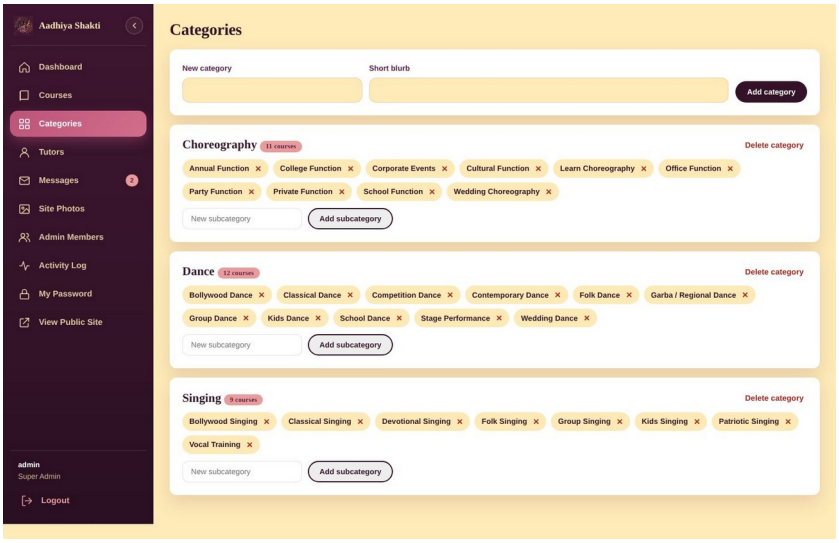
6.14 This page shows: Courses – list with search, category and status filters



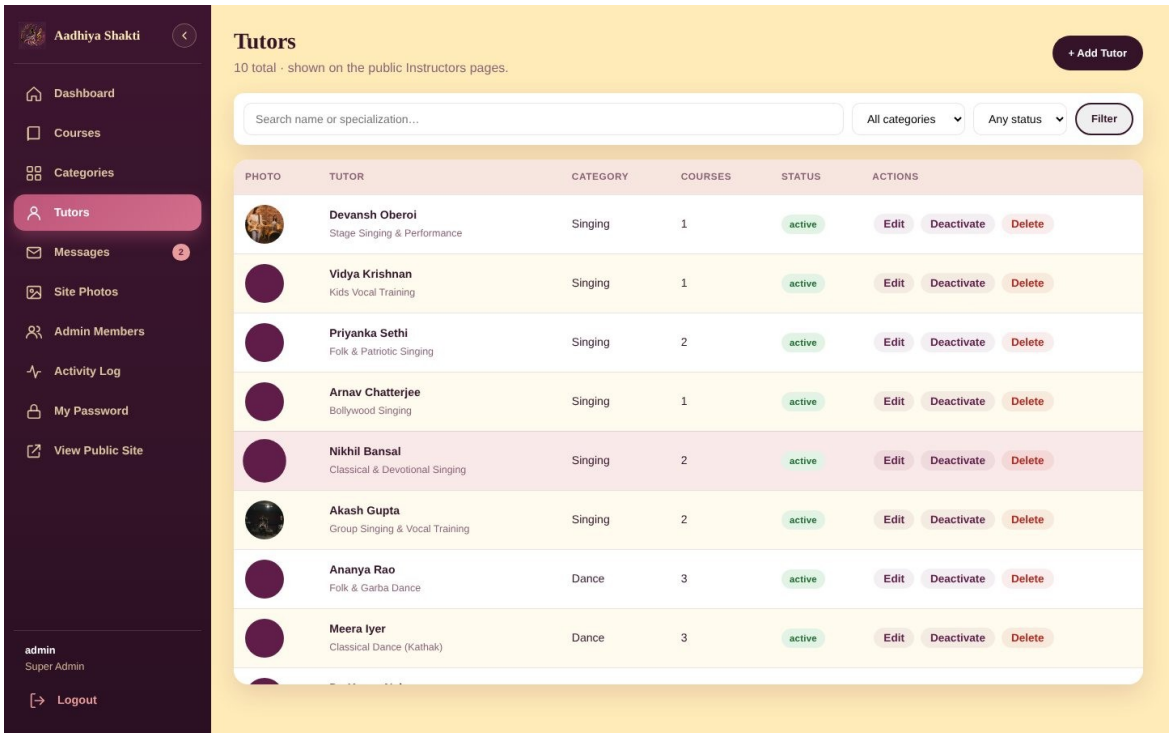
6.15 This page shows: Edit course – category → subcategory selection, tutor, price and image upload with preview



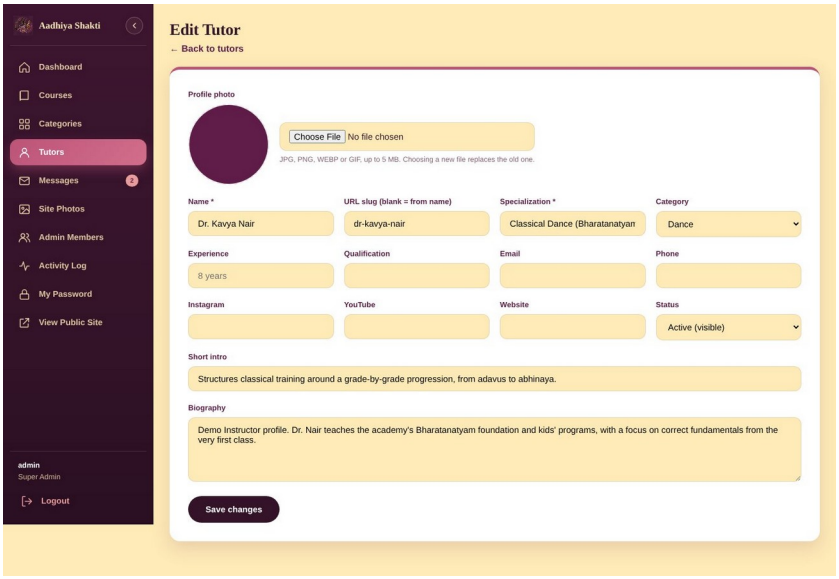
6.16 This page shows: Categories and subcategories management



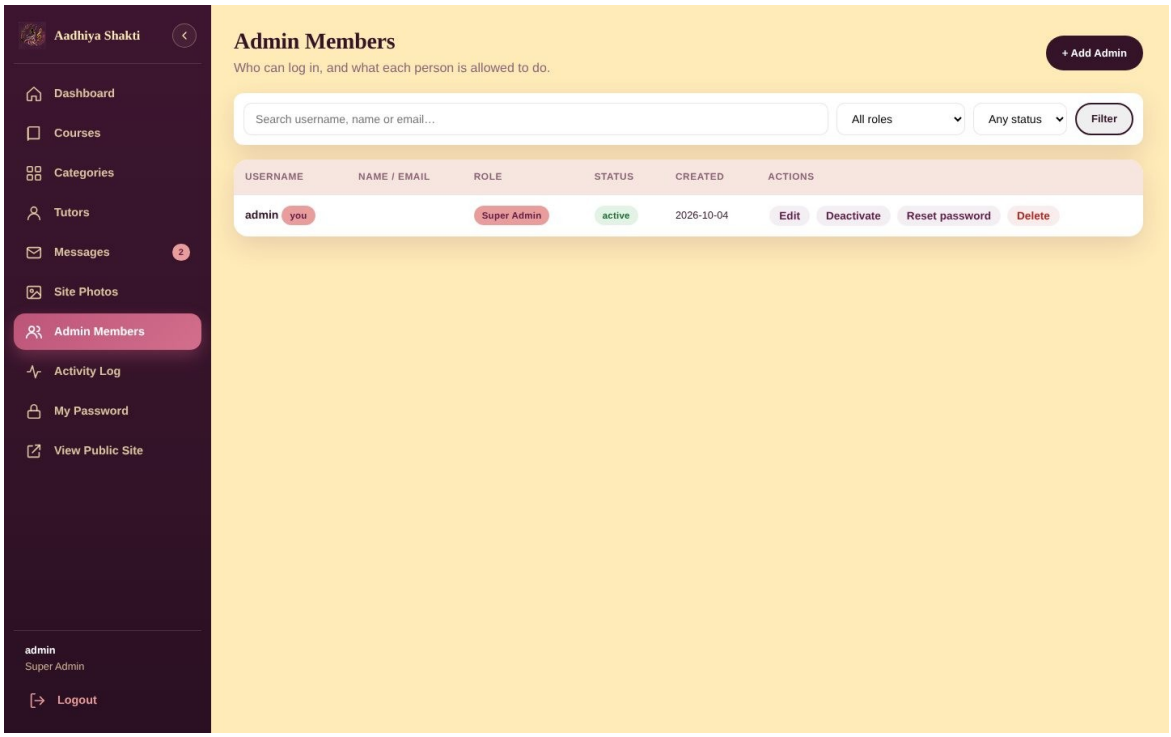
6.17 This page shows: Tutors – list with photo, category, number of courses and status



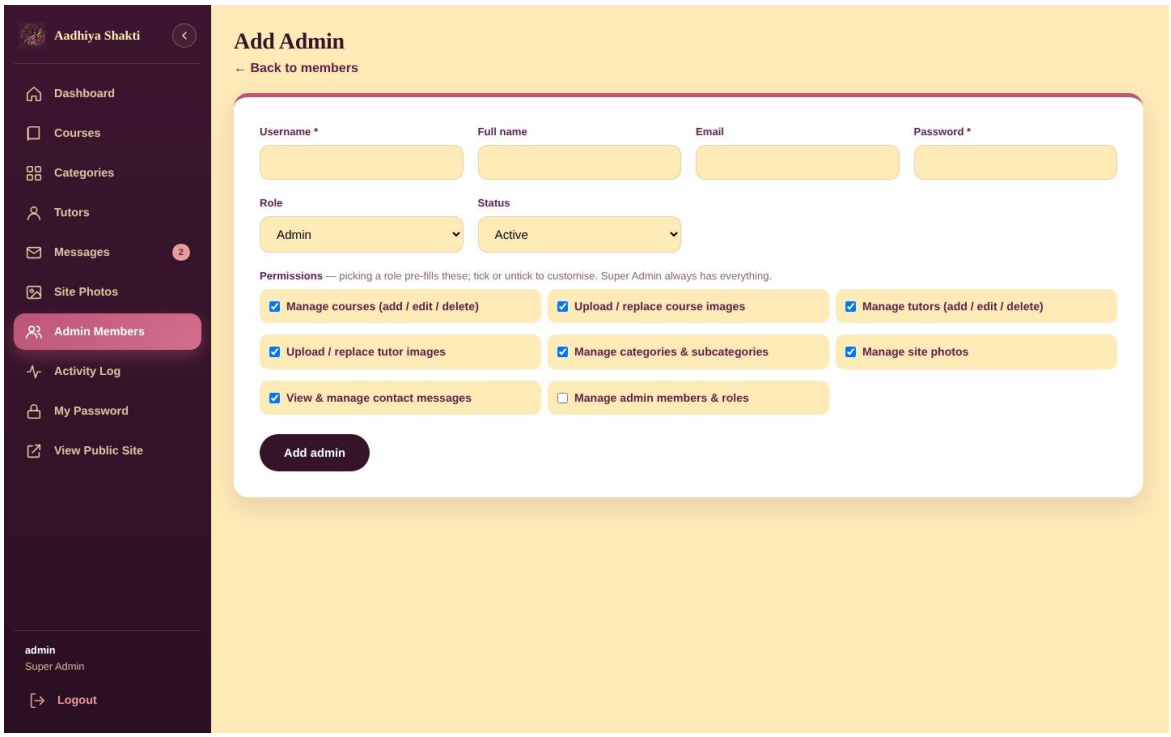
6.18 This page shows: Edit tutor – photo upload with preview, contact and social links



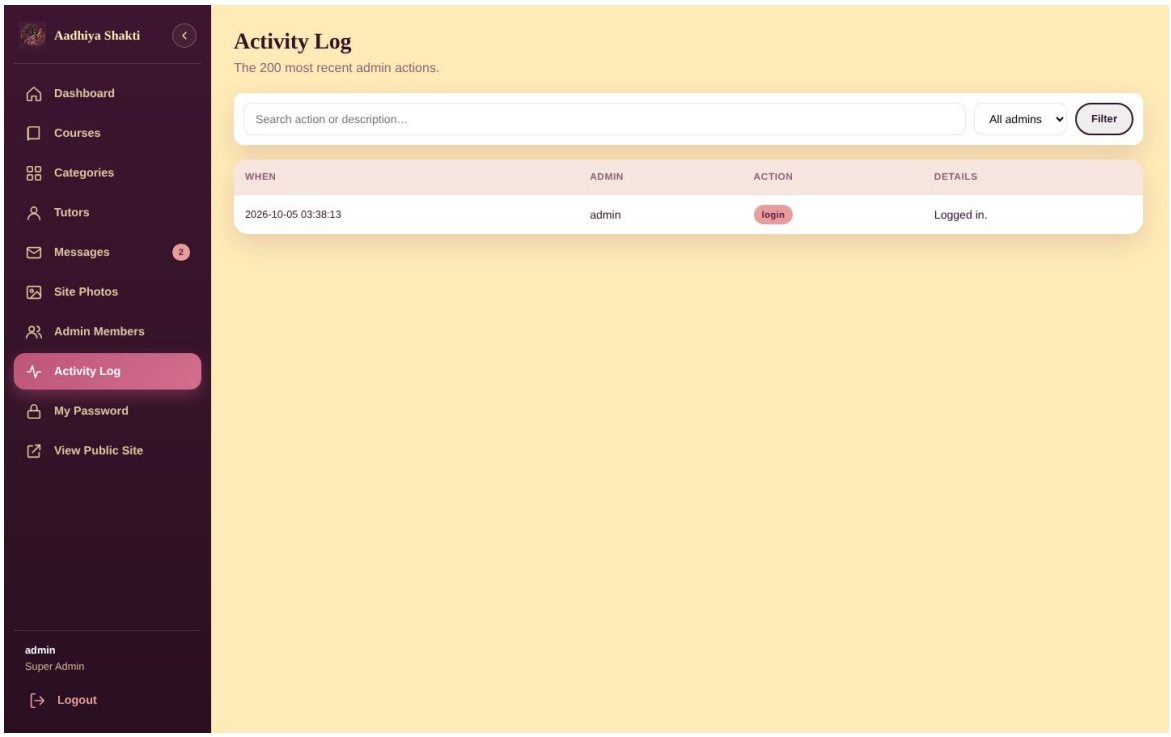
6.19 This page shows: Admin Members – roles, status and actions



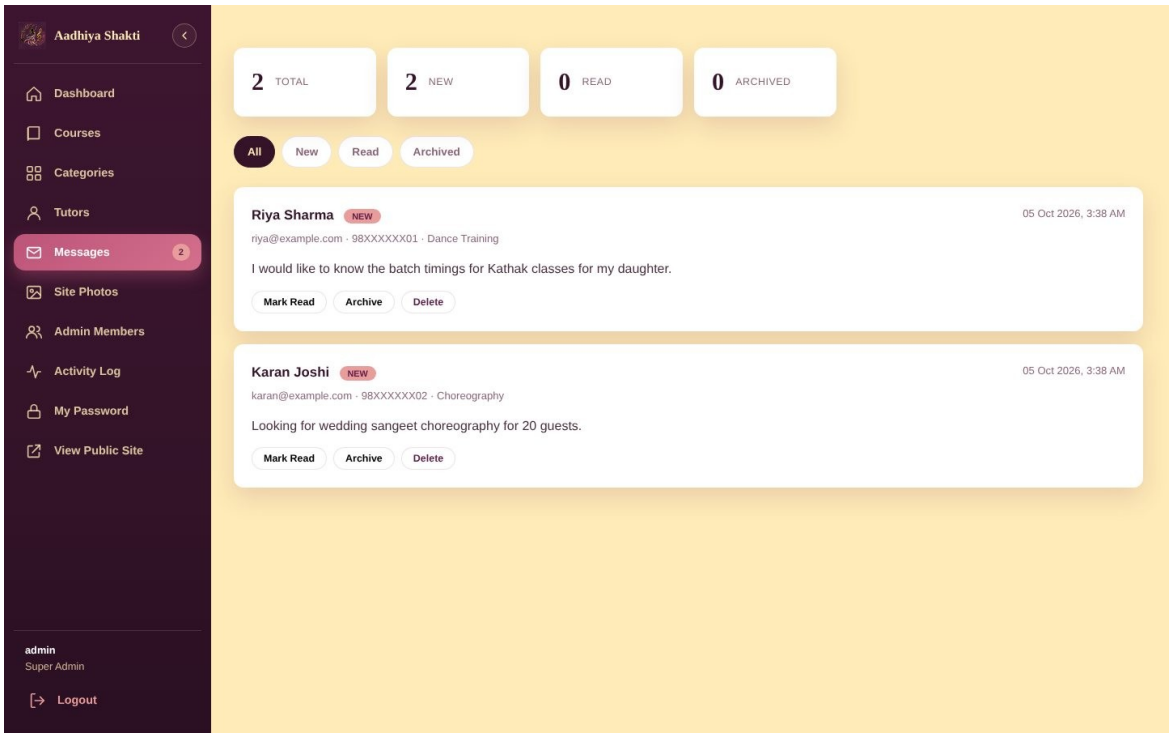
6.20 This page shows: Add admin – role selection with custom permission checkboxes



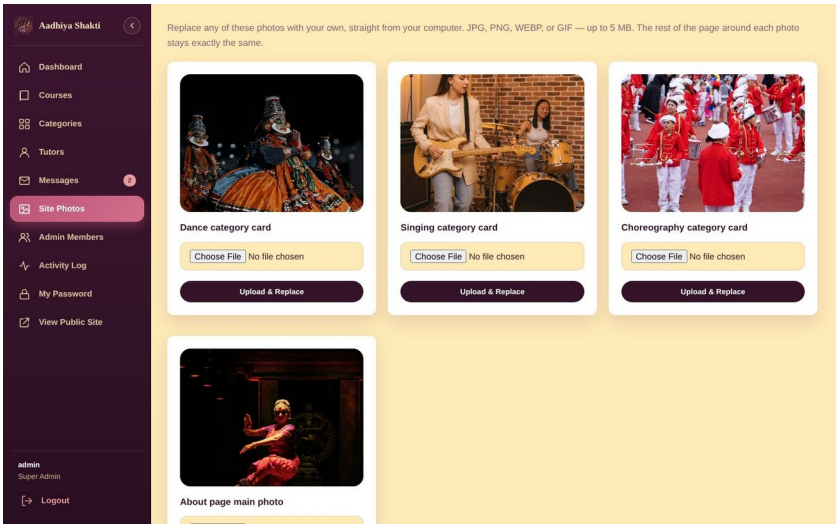
6.21 This page shows: Activity Log – search and filter by admin



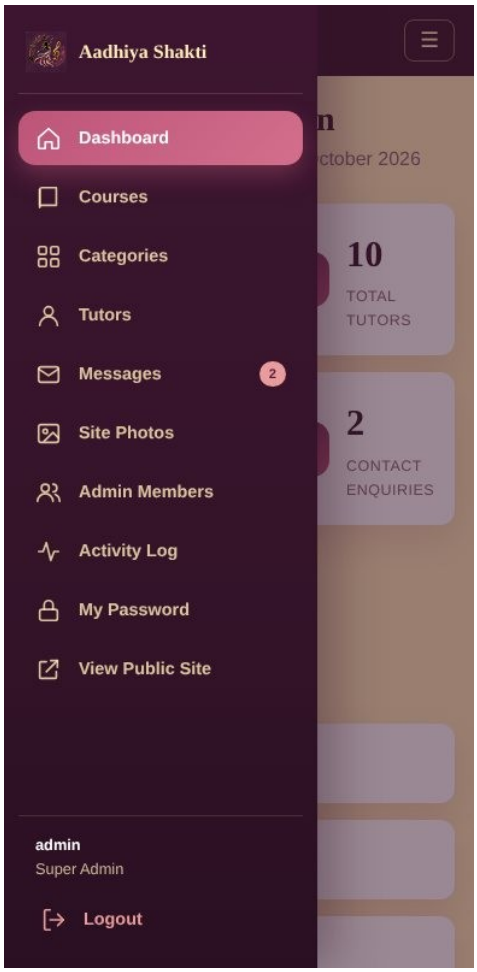
6.22 This page shows: Messages – contact enquiries with status filters



6.23 This page shows: Site Photos – replace category and About photos from the computer



6.24 This page shows: Admin portal on a mobile phone – the sidebar becomes a slide-in drawer



CHAPTER 7

LIMITATIONS AND FUTURE ENHANCEMENTS

Every software system is developed within a defined scope and time. The current limitations of the project and the improvements planned for future versions are described below.

7.1 Limitations

- **No online payment** – prices are shown for information only; fees are collected by the academy directly.
- **No automatic e-mail** – enquiries are stored in the database, but a confirmation e-mail is not yet sent to the visitor.
- **Fixed gallery** – the Gallery page uses a prepared set of images; there is no gallery manager in the admin yet.
- **No student or tutor login** – class booking, attendance and progress tracking are not available.
- **Sample content** – tutor profiles are demonstration profiles, and some contact details are placeholders to be confirmed.
- **Single language** – the website is available in English only.
- **Internet dependency** – some hosted photographs and fonts need an internet connection.

7.2 Future Enhancements

- **Gallery manager** – add, edit, replace, delete and activate gallery images from the Admin Portal.
- **AJAX contact form with e-mail** – instant confirmation on screen and an automatic e-mail through secure SMTP, with credentials stored only on the server.
- **Advanced animations** – glowing animated borders for cards and a timeline animation for events.
- **Online booking** – trial-class requests, class booking and a student dashboard.
- **Payment gateway** – optional online fee payment in a later version, if the academy decides to accept it.
- **Multi-language support** – Gujarati and Hindi versions of the website.
- **Reports** – monthly enquiries and popular courses, with export to Excel.
- **Stronger security** – two-factor authentication for Super Admin accounts.





CHAPTER 8

CONCLUSION

The Aadhiya Shakti Academy project has successfully achieved its objective of providing a dance, singing, music and choreography academy with a professional online presence and a secure, easy-to-use Admin Portal. Visitors can explore courses, tutors, photographs and events and contact the academy, while the administrator can manage courses, categories, tutors, photos, enquiries and admin accounts without touching the code.

During this project we gained practical knowledge of the complete software development life cycle. We learnt how to design a normalised MySQL database, write secure PHP using PDO and prepared statements, protect forms with CSRF tokens, store passwords as hashes, validate file uploads and build role-based access control that is enforced on the server.

The project also taught us the value of working in phases, testing every feature and documenting the work clearly. Its modular structure allows new features such as a gallery manager, e-mail confirmation and online booking to be added easily. We believe this project is a practical and complete example of a modern database-driven web application.



CHAPTER 9

REFERENCES AND BIBLIOGRAPHY

Websites

1. PHP Manual – https://www.php.net/manual/en/
2. PHP: PDO – https://www.php.net/manual/en/book.pdo.php
3. PHP: password_hash – https://www.php.net/manual/en/function.password-hash.php
4. MySQL Reference Manual – https://dev.mysql.com/doc/
5. MAMP – https://www.mamp.info/
6. MDN Web Docs (HTML, CSS, JavaScript, IntersectionObserver) – https://developer.mozilla.org/
7. W3Schools – https://www.w3schools.com/
8. OWASP Top 10 and Cheat Sheet Series – https://owasp.org/
9. Google Fonts (Fraunces, Manrope) – https://fonts.google.com/
10. Unsplash – https://unsplash.com/ and Pexels – https://www.pexels.com/ (photographs, used under their licences)
11. Graphviz – https://graphviz.org/ (diagram generation)

Books

1. Ian Sommerville – Software Engineering, Pearson
2. Roger S. Pressman – Software Engineering: A Practitioner's Approach, McGraw-Hill
3. Robin Nixon – Learning PHP, MySQL & JavaScript, O'Reilly
4. Elmasri & Navathe – Fundamentals of Database Systems, Pearson



APPENDIX A

INSTALLATION AND SETUP GUIDE

1. Install and start **MAMP**; both Apache and MySQL lights must be green.
2. Unzip the project and copy the folder **static-site** into **/Applications/MAMP/htdocs/**.
3. Open **http://localhost:8888/phpMyAdmin**. Click **Import**, choose **database/full-setup.sql** and press **Go**. The file creates the database **aadhiya_shakti_site** with all nine tables, categories, courses, tutors and the first admin.
4. Check **includes/config.php**: DB_HOST 127.0.0.1, DB_PORT 8889, DB_NAME aadhiya_shakti_site, DB_USER root, DB_PASS root (MAMP defaults).
5. Open **http://localhost:8888/static-site/index.php** for the public site.
6. Open **http://localhost:8888/static-site/admin/login.php** and log in with **admin / ChangeMe123!**. Change the password immediately from “My Password”.
7. Add other admins from **Admin Members → Add Admin** and choose a role.

Common problems

Message	Cause and solution
Access denied for user	User or password in config.php does not match MAMP MySQL.
Unknown database	full-setup.sql was not imported or the name differs.
Connection refused	MySQL not started or the port is not 8889.
Images not showing	Check internet (hosted photos) and the uploads folder permission.



APPENDIX B

IMPORTANT SOURCE CODE

1. Database connection (includes/config.php)

```
function db(): PDO
{
    static $pdo = null;
    if ($pdo === null) {
        $pdo = new PDO(
            'mysql:host=' . DB_HOST . ';port=' . DB_PORT . ';dbname=' . DB_NAME .
';charset=utf8mb4',
            DB_USER, DB_PASS,
            [PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
             PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
             PDO::ATTR_EMULATE_PREPARES => false]
        );
    }
    return $pdo;
}
```

2. Permission check enforced on the server

```
function admin_can(string $permission): bool
{
    if (($_SESSION['admin_role'] ?? '') === 'super_admin') return true;
    // otherwise read the admin's saved permission list (JSON) from admin_users
    ...
    return in_array($permission, $cache, true);
}

function admin_require_permission(string $permission): void
{
    if (admin_can($permission)) return;
    http_response_code(403);      // show "Access denied" page
    ...
    exit;
}
```

3. Secure image upload helper

```
function save_uploaded_image(string $field, string $prefix): array
{
    $file = $_FILES[$field] ?? null;
    if ($file['size'] > 5 * 1024 * 1024) return ['error' => 'Image is larger than 5
MB.'];
    $info  = @getimagesize($file['tmp_name']);          // checks REAL image content
    $types = [IMAGETYPE_JPEG => 'jpg', IMAGETYPE_PNG => 'png',
              IMAGETYPE_WEBP => 'webp', IMAGETYPE_GIF => 'gif'];
    if ($info === false || !isset($types[$info[2]])) return ['error' => 'Not a valid
image.'];
    $name = $prefix . '-' . bin2hex(random_bytes(6)) . '.' . $types[$info[2]];  // random
name
    move_uploaded_file($file['tmp_name'], $dir . $name);
    return ['path' => 'assets/images/uploads/' . $name];
}
```



4. Admin logout (admin/logout.php)

```
admin_log('logout', 'Logged out.');
$_SESSION = [];
setcookie(session_name(), '', time() - 42000, ...);   // expire the cookie
session_destroy();
redirect('../index.php');                              // public home page
```

5. Safe tutor deletion (admin/tutors.php)

```
$assigned = count of courses where instructor_slug = tutor.slug
if ($assigned > 0) {
    // must choose another tutor (or "no tutor") first
    if ($to === '__ask__') redirect('tutors.php?reassign=' . $id);
    UPDATE courses SET instructor_slug = ? WHERE instructor_slug = ?
}
DELETE FROM tutors WHERE id = ?
```